The Moduli Space of Formal Systems: Classification, Stabilization, and a No-Go Theorem for Absolute Foundations

Max Sereda
Independent researcher
max@gst.st

August 21, 2026

Abstract

I study the (∞, n)-categorical moduli space $\mathfrak{M}$ of formal systems (hereafter **MSFS**): the classifying 2-stack of Morita-equivalence classes of Rich-foundations ($\mathcal{L}_{\mathrm{Fnd}}$ theories satisfying (R1)–(R5)). I establish five structural results and one boundary corollary.

(i) Plurality at $\mathcal{L}_{\mathrm{Cls}}$. The ∞-cosmoi (Riehl–Verity), Univalent Foundations (Voevodsky), and cohesive higher topoi (Schreiber) are pairwise non-2-equivalent partial meta-frameworks, each classifying a strict sub-stack of $\mathfrak{M}$; they lie outside the rigid maximal sub-class $\mathfrak{Meta}^{\top}_{\mathrm{Cls}}$.

(ii) Conditional meta-categoricity. Any two members of $\mathfrak{Meta}^{\top}_{\mathrm{Cls}}$ over the same Rich-metatheory are (∞, ∞)-equivalent, by Grothendieck–Lurie straightening with joint-faithfulness of the extensional and intensional classification functors.

(iii) Slice-local intensional refinement. The display-2-class functor $\mathbf{I} : \mathcal{F}^{\mathrm{op}} \to \mathcal{S}_{\mathrm{int}}$ is slice-local over $\mathfrak{M}$: intensional distinctions (MLTT vs. ETT) fibre over single $\mathfrak{M}$-points, invariantly separated by internal-language computability in Hyland's effective topos Eff.

(iv) Theory-level meta-stabilization. Iterated meta-classification reproduces the same (∞, ∞)-theory at each step (Barwick–Schommer-Pries unicity), though set-theoretic instantiation ascends the Grothendieck-universe hierarchy $\kappa_1 < \kappa_2 < \cdots$.

(v) AC/OC duality. The Lambek–Scott adjunction $(\mathrm{Syn}, \mathrm{Mod})$ of (R5b), assembled uniformly across R-S and over $n \in \mathbb{N} \cup \{\infty\}$, induces an (∞, ∞)-equivalence $\mathfrak{M} \simeq \mathfrak{M}_{\mathcal{E}}$ between the theory-centric moduli 2-stack $\mathfrak{M}$ and an enactment-centric moduli 2-stack $\mathfrak{M}_{\mathcal{E}}$ of pointed Rich-enactments. The dual of AFN-T asserts that the enactment-absolute stratum is likewise empty.

Boundary corollary: the Absolute Foundation No-Go Theorem (AFN-T). By the syntax–semantics adjunction underlying (R5), $\mathfrak{M}$ has no maximal point: the $\mathcal{L}_{\mathrm{Abs}}$ stratum (formally definable, non-reducible, and maximally generative foundation) is empty. AFN-T subsumes the classical no-go series Cantor [10], Russell [43], Gödel [17], Tarski [47], Lawvere [35], Ernst [12] uniformly across R-S and all (∞, n), $n \in \mathbb{N} \cup \{\infty\}$, and extends dually to the enactment-absolute stratum $\mathcal{L}^{\mathcal{E}}_{\mathrm{Abs}}$.

Keywords: moduli space of formal systems, classifying 2-stacks, categorical logic, (∞, n)-categories, higher topos theory, intensional type theory, meta-classification, effective topos, no-go theorems.

MSC2020: 03A05, 03B30, 03C45, 03F40, 18A05, 18N10, 18N50, 18N60, 18N65.

Contents

1 Introduction

This paper studies the *moduli space of formal systems* $\mathfrak{M}$, the classifying $(\infty, 2)$-stack whose points are Morita-equivalence classes of Rich-foundations (formal theories F satisfying conditions (R1)–(R5) of Definition 3.1) and whose k-morphisms are faithful interpretations and provable natural equivalences. The principal results concern the structure of $\mathfrak{M}$:

- **Stratification** (§2): $\mathcal{L}_{\mathrm{Fnd}} \xrightarrow{\mathrm{Cls}} \mathcal{L}_{\mathrm{Cls}} \supsetneq \mathcal{L}_{\mathrm{Cls}}^{\top} \xrightarrow{\mathrm{Gen}} \mathcal{L}_{\mathrm{Abs}} = \emptyset$ — the strata are cut out by type-distinct condition packages ((R1)–(R5) on formal systems vs. (M1)–(M5) on classifiers, Proposition 2.2 (ii)); the sole inclusion is $\mathcal{L}_{\mathrm{Cls}}^{\top} \subsetneq \mathcal{L}_{\mathrm{Cls}}$.
- **Structural plurality at $\mathcal{L}_{\mathrm{Cls}}$** (Theorem 9.4).
- **Conditional categoricity of $\mathcal{L}_{\mathrm{Cls}}^{\top}$** (Theorem 9.3).
- **Theory-level stabilization with universe-ascent** (Theorem 9.6).
- **Slice-locality of intensional refinement** (Theorem 8.7).
- **Intensional grading of the slice fibers** (Definition 8.9, Proposition 8.10): each fiber $\mathrm{Int}([F])$ carries a canonical functorial $\mathbb{N}$-filtration by display grade; extensionality is exactly grade-blindness (Corollary 8.11).
- **Five-axis absoluteness** (§7).
- **AC/OC Morita duality** (Theorem 10.5, Theorem 10.8).
- **Outer boundary:** $\mathcal{L}_{\mathrm{Abs}} = \emptyset$ — the Absolute Foundation No-Go Theorem (Theorem 5.1, Theorem 6.1).

Machine verification. Every theorem, lemma, and corollary in this paper has a companion machine-checked formal proof in the **Verum MSFS Corpus** (`https://github.com/gst-st/msfs`), self-contained modulo ZFC + 2 strongly inaccessible cardinals. See Appendix C for a one-paragraph entry-point and follow the corpus repository for the live audit surface, paper-to-corpus map, and reproducibility instructions.

1.1 Notation and Conventions

Convention 1.1. I work throughout in ZFC augmented by two inaccessible cardinals $\kappa_1 < \kappa_2$, providing Grothendieck universes $\mathbf{U}_1 \subset \mathbf{U}_2$ sufficient for the 2-categorical machinery used. The categorical machinery follows Kelly [33] for enriched and 2-categories, Lurie [36] for $(\infty, 1)$-categories, Lurie [37] for higher algebra, and Riehl–Verity [42] for ∞-cosmoi. For accessible functor theory I follow Adámek–Rosický [2].

Convention 1.2 (Key symbols).

- $\mathcal{F}$ denotes the 2-category of *Rich-foundations* ($\mathcal{L}_{\mathrm{Fnd}}$): objects are foundational theories in the sense formalised in §3 below (conditions (R1)–(R5)); 1-morphisms are faithful interpretations; 2-morphisms are provable natural equivalences between interpretations.
- $\rho : \mathcal{F} \to (\infty, n)\text{-Cat}$ denotes the *classification functor* assigning to each $F \in \mathcal{F}$ its category of models (or, for type-theoretic F, the syntactic (∞, n)-category $\mathrm{Syn}(F)$).

- A *gauge transformation* $\tau : F_1 \to F_2$ is a 1-morphism in $\mathcal{F}$ accompanied by a distinguished 2-equivalence $\rho(F_1) \simeq \rho(F_2)$ at the model-categorical level. Two foundations are *gauge-equivalent* if there is an invertible gauge transformation between them. *Relation to Morita equivalence.* Gauge-equivalence at the $\mathcal{F}$-level corresponds, via the $\mathfrak{M} := \mathcal{F}/\text{gauge}$ quotient, to Morita equivalence at the stack level: $F_1, F_2 \in \mathcal{F}$ are gauge-equivalent iff $[F_1] = [F_2]$ in $\mathfrak{M}$, which by construction is the Morita-equivalence class. Throughout the paper, "gauge" (1-morphism level) and "Morita" (stack level) name the same underlying equivalence relation at their respective categorical strata, related by the $\mathcal{F} \to \mathfrak{M}$ quotient functor; no distinct third relation is introduced.

- *Morita-reducibility at the* (∞, n)*-level*: a $\mathcal{L}_{\text{Abs}}$ candidate X and any $Y \in \mathcal{S}_S$ are each viewed as pointed (∞, n)-categorical structures via the canonical Yoneda embedding into the total (∞, n)-category $\mathbf{StrCat}_{S,n}$ of S-structured (∞, n)-categories (Definition 1.3, stated immediately below). Concretely, X presented by a formula ϕ_X corresponds to the pair $(\text{Syn}(S), [\phi_X])$ where $[\phi_X]$ is the isomorphism class of the object defined by ϕ_X in the syntactic category; $Y \in \text{Ob}(\mathcal{M}) \subseteq \mathcal{S}_S^{\text{local}}$ corresponds to $(\mathcal{M}, Y)$. I write $X \to_{M,n} Y$, and say X is *Morita-reducible at the* (∞, n)*-level* to Y, if there exists an (∞, n)-functor $\Phi : \text{Syn}(S) \to \mathcal{M}$ in $\mathbf{StrCat}_{S,n}$ sending $[\phi_X]$ to the isomorphism class of Y, with the restriction $\Phi|_{[\phi_X]}$ an (∞, n)-equivalence onto its image. This extends classical ring-Morita equivalence (which compares $\text{Mod}(X)$ and $\text{Mod}(Y)$ as 1-categories) by incorporating the full (∞, n)-categorical structure; the two notions coincide when X, Y are 1-categorical and $n = 1$. I use the shorthand "Morita-equivalent" for the symmetric notion and "Morita-reducible" for the directed version.

- $\mathfrak{M}$ denotes the *classifying* 2*-stack* of $\mathcal{F}$: the quotient $\mathfrak{M} := \mathcal{F}/\text{gauge}$, regarded as an $(\infty, 2)$-stack of Morita-equivalence classes of Rich-foundations. Formally, $\mathfrak{M}$ is the straightening of the Cartesian fibration $\int \rho \to \text{R-S}$ whose fiber over S is the 2-groupoid $\rho^{-1}(S)$ of foundations over S modulo gauge. The construction follows Lurie HTT [36] §3.2 (straightening–unstraightening of Cartesian fibrations).

- For a 2-category $\mathbf{F}$, $\text{Aut}_2(\mathbf{F})$ denotes the 2-group of 2-autoequivalences $\mathbf{F} \xrightarrow{\simeq_2} \mathbf{F}$ (monoidal under composition, with invertible 1-morphisms and coherent 2-cells).

- $\mathcal{S}_{\text{int}}$ denotes the 2-category of display 2-categories (rigorous definition deferred to Appendix A below).

- All 2-categories are locally small and κ_1-accessible unless stated otherwise. (For specific Rich-foundations S whose consistency strength requires the second inaccessible, the $\text{Syn}(S)$-accessibility parameter generalizes to $\kappa_S \in \{\kappa_1, \kappa_2\}$ per the accessibility clause (R5a) of the Rich-metatheory definition, formalised in §3 below; subsequent κ_1-statements should be read uniformly as κ_S-statements where S is indexed by the specific $\mathcal{L}_{\text{Fnd}}$-point under consideration. All proofs in this paper are stable under this uniform reading, since the accessibility arguments invoke only regularity of the cardinal, not its specific value.)

- *Truncation and lifting across depths*: for $n \leq n_S$ I write $\text{Syn}^{(\infty,n)}(S) := \tau_{\leq n}(\text{Syn}(S))$ for the (∞, n)-truncation of the syntactic category. For $n > n_S$ I write $\text{Syn}^{(\infty,n)}(S)$ for the canonical (∞, n)-lifting, obtained by left Kan extension of $\text{Syn}(S)$ along the inclusion $(\infty, n_S)\text{-Cat} \hookrightarrow (\infty, n)\text{-Cat}$ (Lurie HTT [36] §5.4). Under this convention, comparisons between an object X at level n and an S-definable object Y at the native depth n_S take place at level $\max(n, n_S)$ via the canonical lifts; when $n = \infty$, the colimit $\text{colim}_{k<\infty} \text{Syn}^{(\infty,k)}(S)$ is the resulting ambient.

Definition 1.3 (The total (∞, n)-category $\mathbf{StrCat}_{S,n}$)**.** For $S \in$ R-S and $n \in \mathbb{N} \cup \{\infty\}$, the *total* (∞, n)*-category of* S*-structured* (∞, n)*-categories*, denoted $\mathbf{StrCat}_{S,n}$, has:

- **Objects**: pairs $(\mathcal{C}, \mathrm{str}_{\mathcal{C}})$ where $\mathcal{C}$ is an (∞, n)-category that is either essentially $\mathbf{U}_1$-small or κ_1-accessible (the two cases cover syntactic categories and model categories respectively; cf. (R5a)) and $\mathrm{str}_{\mathcal{C}}$ is an S-model structure on $\mathcal{C}$ (i.e., an interpretation of S-signature into $\mathcal{C}$ satisfying the axioms of S).

- **1-morphisms**: S-preserving (∞, n)-functors $\Phi : (\mathcal{C}_1, \mathrm{str}_1) \to (\mathcal{C}_2, \mathrm{str}_2)$, i.e., functors commuting with the respective S-model structures up to coherent (∞, n)-isomorphism.

- **Higher morphisms**: coherent (∞, n)-natural transformations between S-preserving functors.

I embed objects of $\mathrm{Syn}(S)$ and of any model $\mathcal{M} \models S$ into $\mathbf{StrCat}_{S,n}$ via the pointing construction $X \mapsto (\mathrm{Syn}(S), X)$ and $Y \mapsto (\mathcal{M}, Y)$ respectively. Morita-reducibility, as specified in Convention 1.2, refers to the existence of an S-preserving (∞, n)-functor in $\mathbf{StrCat}_{S,n}$ between such pointed structures that is an equivalence onto its image.

1.2 Assumption Register

Every theorem below is proved *from* a specific package of assumptions, and the paper's headline claims ("$\mathcal{L}_{\mathrm{Abs}} = \emptyset$", "$\mathfrak{Meta}^{\top}_{\mathrm{Cls}}$ is categorical") are exactly as strong as that package. Because downstream work builds on MSFS as a base — in particular, any extension that positions itself as a *limiting* formal theory must either discharge or explicitly inherit each entry — I collect the assumptions here in one register, each with its formal status. Nothing in this subsection is new mathematics; it is the paper's trusted surface made enumerable.

(A1) Foundational base. ZFC + two strongly inaccessible cardinals $\kappa_1 < \kappa_2$ (Convention 1.1). *Status: foundational axiom package.* All results are theorems of this metatheory; by Gödel II the package cannot certify its own consistency, and the paper nowhere claims otherwise (cf. open question (Q4), §12). Consistency-strength statements downstream can at best be equiconsistencies with (A1).

(A2) Scope of "metatheory". The no-go quantifies over R-S as delimited by (R1)–(R5) (§3): recursively enumerable, interpreting Robinson's Q, possessing a model in $\mathbf{U}_2$, $\mathbf{StrCat}$-compatible, with essentially $\mathbf{U}_1$-small $\mathrm{Syn}(S)$ of κ_S-accessible Ind-completion (R5a). *Status: definitional scope.* Systems outside R-S — non-r.e. oracle theories, infinitary logics without accessible syntax — are not ruled on. Theorem 5.1 is universal over R-S, not over an informal totality of "all foundations".

(A3) Formal content of "absolute foundation". $\mathcal{L}_{\mathrm{Abs}}$ membership is *defined* as the conjunction $(F_S) \wedge (\Pi_{4,S,n}) \wedge (\Pi_{3\text{-max},S,n})$ (Definition 4.4). *Status: interpretive adequacy assumption.* AFN-T forecloses precisely this triple; the philosophical reading "no absolute foundation exists" is faithful exactly to the extent the triple captures the intended notion. The five-axis analysis of §7 hardens this reading axis-by-axis but does not eliminate its interpretive character.

(A4) Comparison relation. Objects are compared through Morita-reducibility: existence of an S-preserving (∞, n)-functor in $\mathbf{StrCat}_{S,n}$ that is an equivalence onto its image, with objects embedded by the pointing construction (Definition 1.3). *Status: structural convention.* A finer or coarser comparison topology would redraw stratum boundaries; §8.1 measures exactly what the Morita quotient forgets.

(A5) Axis exhaustiveness. The escape-parameter space is taken to be the five axes (S, n, μ, ξ, π) of §7. *Status: definitional scope, not a theorem.* Exhaustiveness holds by definition of what the paper counts as an escape parameter; a genuinely new axis would require a new theorem, not contradict an existing one.

(A6) Meta-classifier axioms. The classifying apparatus satisfies (M1)–(M5) and, for the maximal class, (Max-1)–(Max-4) — including the joint-conservativity clause of (Max-4) on the pair $(\mathrm{Cl}_{\mathbf{F}}, \mathbb{I}_{\mathbf{F}})$, which powers the faithfulness step of Theorem 9.3. *Status: axioms on the classifier, motivated but not derived.*

(A7) Localization semantics of equivalence. The AC/OC duality (§10) reads "equivalence" through $\mathrm{gauge}_{\mathcal{E}}$, the 2-localization of $\mathcal{E}$ at the reflection morphisms $\mathcal{R}_{\mathcal{E}}$ (Pronk calculus of fractions, (BF1)–(BF5)): the reflection unit is inverted *by* the localization, never in $\mathcal{E}$ itself. *Status: semantic convention*, of the same kind as ring-theoretic Morita equivalence.

(A8) Minimal display class. Intensional structure is measured against the *canonical minimal* display class $\mathcal{D}_{\mathbf{F}}$, inductively generated by (D1)–(D4), and the display grade of §8.1 is relative to this generation. *Status: canonical-minimality convention;* a larger display class would coarsen the grading but not disturb the extensional quotient.

Remark 1.4 (Interface for limiting extensions)**.** The register is designed as a formal interface. An extension of MSFS claiming any form of *limit* status can be audited entry-by-entry: (A1) can only be inherited (equiconsistency is the ceiling); (A2) and (A5) can be *internalized* — proved equivalent to structural trade-offs inside the extension — but not escaped; (A3), (A4), (A7), (A8) are conventions an extension must either adopt verbatim or re-justify; (A6) may be strengthened. In particular, a limiting extension is consistent with AFN-T precisely when its limit claim is indexed by this register — maximality *within* $\mathcal{L}_{\mathrm{Cls}}^{\top}$-type strata — rather than membership in $\mathcal{L}_{\mathrm{Abs}}$, which Theorem 5.1 leaves empty.

2 Stratified Hierarchy of the Moduli Space

I stratify $\mathfrak{M}$ into four formal strata distinguished by the two structurally distinct meta-operations Cls (horizontal, classifying) and Gen (vertical, generative). The strata, defined by their categorical conditions, are:

$$\mathcal{L}_{\mathrm{Fnd}} \xrightarrow{\mathrm{Cls}} \mathcal{L}_{\mathrm{Cls}} \supsetneq \mathcal{L}_{\mathrm{Cls}}^{\top} \xrightarrow{\mathrm{Gen}} \mathcal{L}_{\mathrm{Abs}}.$$

2.1 The Formal Strata

Definition 2.1 (Stratified hierarchy — formal)**.** I define the formal *strata* as classes of mathematical objects:

Class	Membership criterion
$\mathcal{L}_{\mathrm{Fnd}}$	Formal systems satisfying (R1)–(R5) (specified in §3 below)
$\mathcal{L}_{\mathrm{Cls}}$	Classifiers satisfying (M1)–(M5) (specified in §9 below)
$\mathcal{L}_{\mathrm{Cls}}^{\top}$	Classifiers additionally satisfying (Max-1)–(Max-4) (specified in §9 below)
$\mathcal{L}_{\mathrm{Abs}}$	Objects satisfying $(F_S) \wedge (\Pi_{4,S,n}) \wedge (\Pi_{3\text{-max},S,n})$ for some (S, n) (specified in §4 below)

Proposition 2.2 (Structural properties of the hierarchy)**.** ***(i)*** Definability. *Each of the four strata is a (possibly proper) class definable in* $\mathsf{ZFC} + 2$*-inacc augmented by the 2-categorical*

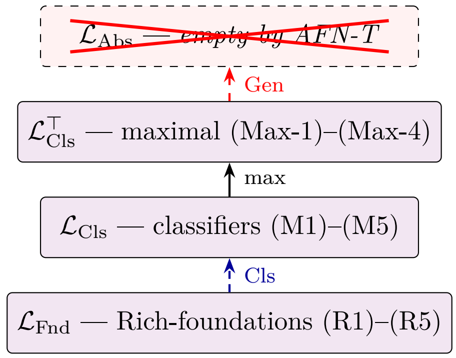

Figure 1: The formal stratification of $\mathfrak{M}$. The blue dashed arrow Cls denotes the *horizontal meta-operation* (classification without generation; see §2.3), taking $\mathcal{L}_{\mathrm{Fnd}}$ to $\mathcal{L}_{\mathrm{Cls}}$. The maximality restriction selects the sub-class $\mathcal{L}^{\top}_{\mathrm{Cls}}$. The red dashed arrow Gen denotes the *vertical meta-operation* (maximal generativity), whose image $\mathcal{L}_{\mathrm{Abs}}$ is empty by AFN-T (Theorem 5.1). $\mathcal{L}_{\mathrm{Fnd}}$ includes ZFC, MLTT, HoTT, $(\infty,1)$-topos theory, and noncommutative geometry. $\mathcal{L}_{\mathrm{Cls}}$ partitions into the maximal sub-class $\mathfrak{Meta}^{\top}_{\mathrm{Cls}}$ (categorical by Theorem 9.3) and a plural complement of *partial* classifiers (Theorem 9.4; ∞-cosmoi, Univalent Foundations, cohesive $(\infty,1)$-topoi), each classifying a strict sub-stack of $\mathfrak{M}$. By Theorem 10.5, the diagram applies verbatim to the enactment-centric moduli stack $\mathfrak{M}_{\mathcal{E}}$ via the (∞,∞)-equivalence $\mathfrak{M} \simeq \mathfrak{M}_{\mathcal{E}}$: the dual strata $\mathcal{L}^{\mathcal{E}}_{\mathrm{Fnd}} \to \mathcal{L}^{\mathcal{E}}_{\mathrm{Cls}} \supsetneq (\mathcal{L}^{\top}_{\mathrm{Cls}})^{\mathcal{E}} \to \mathcal{L}^{\mathcal{E}}_{\mathrm{Abs}}$ mirror the strata above, with $\mathcal{L}^{\mathcal{E}}_{\mathrm{Abs}}$ likewise empty (Theorem 10.8).

machinery of Convention 1.1. For $\mathcal{L}_{\mathrm{Fnd}}$ the conditions are (R1)–(R5) (§3 below); for $\mathcal{L}_{\mathrm{Cls}}$ they are (M1)–(M5) (§9 below); for $\mathcal{L}^{\top}_{\mathrm{Cls}}$ they strengthen to (Max-1)–(Max-4) (§9 below); $\mathcal{L}_{\mathrm{Abs}}$ is defined by the triple $(F_S) \wedge (\Pi_{4,S,n}) \wedge (\Pi_{3\text{-}max,S,n})$ for some (S,n) (§4 below; the structural self-contradiction of this triple, recorded in §5, yields Theorem 5.1).

(ii) Multi-stratum membership. *$\mathcal{L}_{\mathrm{Fnd}}$ and $\mathcal{L}_{\mathrm{Cls}}$ are characterized by logically different condition sets (first-order $(R1)$–$(R5)$ on a formal system vs. 2-categorical $(M1)$–$(M5)$ on a 2-category). A mathematical object carrying both structures satisfies both sets simultaneously: e.g., Univalent Foundations belongs to $\mathcal{L}_{\mathrm{Cls}}$, with its underlying* HoTT *in $\mathcal{L}_{\mathrm{Fnd}}$. I write* $\mathrm{Stratum}(X) \subseteq \{\mathcal{L}_{\mathrm{Fnd}}, \mathcal{L}_{\mathrm{Cls}}, \mathcal{L}^{\top}_{\mathrm{Cls}}, \mathcal{L}_{\mathrm{Abs}}\}$ *for the set of formal strata containing X.*

(iii) Strict inclusion at the maximal layer. $\mathcal{L}^{\top}_{\mathrm{Cls}} \subsetneq \mathcal{L}_{\mathrm{Cls}}$. *Theorem 9.3 asserts that any two members of $\mathcal{L}^{\top}_{\mathrm{Cls}}$ parametrized by the same Rich-metatheory are (∞,∞)-equivalent. Theorem 9.4 produces ≥ 3 pairwise non-2-equivalent members of the complement $\mathcal{L}_{\mathrm{Cls}} \setminus \mathcal{L}^{\top}_{\mathrm{Cls}}$: the ∞-cosmoi of Riehl–Verity, the Univalent Foundations programme, and the cohesive framework of Schreiber.*

(iv) Emptiness of the absolute stratum. $\mathcal{L}_{\mathrm{Abs}} = \emptyset$ *by Theorem 5.1. Consequently* $\mathcal{L}_{\mathrm{Fnd}} \cap \mathcal{L}_{\mathrm{Abs}} = \mathcal{L}_{\mathrm{Cls}} \cap \mathcal{L}_{\mathrm{Abs}} = \mathcal{L}^{\top}_{\mathrm{Cls}} \cap \mathcal{L}_{\mathrm{Abs}} = \emptyset$ *trivially.*

Proof. **(i)** Each criterion is first-order-expressible; the specified classes are definable.

(ii) The condition sets concern disjoint aspects of an object: axioms with proof calculus vs. 2-categorical structure with classifying functor. An object carrying both structures satisfies both criteria simultaneously.

(iii) Inclusion from the two sets of conditions specified in §9 below: (Max-1)–(Max-4) strengthen (M1)–(M5). Strictness from the structural multiplicity theorem (Theorem 9.4, proved in §9 below): $\mathbf{F}_{\mathrm{cosmoi}}, \mathbf{F}_{\mathrm{univalent}}, \mathbf{F}_{\mathrm{cohesive}}$ lie in $\mathcal{L}_{\mathrm{Cls}} \setminus \mathcal{L}^{\top}_{\mathrm{Cls}}$ since their classification images are strict sub-stacks of $\mathfrak{M}$, violating (Max-1).

(iv) Immediate from Theorem 5.1. $\square$

2.2 Examples

Stratum $\mathcal{L}_{\mathrm{Fnd}}$. Foundations, with characteristic features:

Foundation	Year	Originator	Characteristic
Z (Zermelo)	1908	Zermelo	No Replacement
ZF/ZFC	1922	Zermelo, Fraenkel	Classical set theory
ZFC + inacc	1963+	Grothendieck	With universes
NBG	1925–40	von Neumann, Bernays, Gödel	First-class classes
MK	1955	Morse, Kelley	Impredicative classes
KP/KPU	1965	Kripke–Platek	Admissible sets
ETCS	1964	Lawvere	Elementary category of sets
ETCC	1966	Lawvere	Elementary theory of categories
NF; NFU	1937; 1969	Quine; Jensen	Stratified comprehension
CZF	1978	Aczel	Constructive ZF
IZF	1970s	Friedman	Intuitionistic ZF
PA	1889	Peano	First-order arithmetic
Z_2	20c	Hilbert–Bernays	Second-order arithmetic
MLTT	1984	Martin-Löf	Dependent types
CIC	1988–90	Coquand–Huet; Coquand–Paulin	Inductive constructions
HoTT	2005+	Awodey, Voevodsky	Univalence + identity types
Cubical HoTT	2015+	Cohen–Coquand et al.	Computational univalence
Universe-poly. HoTT	2013+	Voevodsky	Universe polymorphism
Linear logic + !	1987	Girard	Resource-sensitive
NBG + AFA	1988	Aczel	Non-well-founded sets
$(\infty, 1)$-topos theory	2009	Lurie	$(\infty, 1)$-topoi
Non-commutative geom.	1980s	Connes	Spectral triples
Cohesive $(\infty, 1)$-topos	2013	Schreiber	Cohesion modalities
Motivic SH(k)	1999	Voevodsky, Morel	Motivic homotopy
Realizability (Eff)	1982	Hyland	Effective topos
Markov constructive math	1950s–1980s	A.A. Markov, Nagornyj	Constructive analysis + Markov's principle
Bishop constructive analysis	1967	Bishop, Bridges	Choice-free constructive real analysis
Synthetic diff. geom.	1970s	Kock, Lawvere	Infinitesimals as axioms
Elementary higher topos	2021	Rasekh	Elementary $(\infty, 1)$-topos

Stratum $\mathcal{L}_{\mathrm{Cls}}$. Classifiers (meta-frameworks). $\mathcal{L}_{\mathrm{Cls}}$ partitions into the maximal sub-class $\mathfrak{Meta}^{\top}_{\mathrm{Cls}}$ (categorical up to (∞, ∞)-equivalence by Theorem 9.3) and its complement of *partial* (non-maximal) classifiers, which classify strict sub-stacks of $\mathfrak{M}$ rather than all of $\mathfrak{M}$. The classifiers listed below are all non-maximal: each satisfies (M1)–(M5) but violates (Max-1), as witnessed by the strict-subset character of its classification focus.

Meta-framework	Year	Originator	Classification focus	In $\mathfrak{Meta}^{\top}_{\mathrm{Cls}}$?
∞-cosmoi	2022	Riehl–Verity	$(\infty,1)$-categorical theories	No
Univalent Foundations	2010+	Voevodsky	HoTT extensions	No
Cohesive framework	2013	Schreiber	Cohesive ∞-topoi	No
Higher Algebra prog.	2017+	Lurie	Stable ∞-cat + operadic	No
Synthetic mathematics	2000s+	Taylor, Shulman	Axiomatic synthetic foundations	No

Stratum $\mathcal{L}_{\mathrm{Abs}}$. Empty, by AFN-T.

2.3 The Two Meta-Operations

Horizontal and vertical meta. Two categorically distinct meta-operations act on a stratum $\mathcal{L}$:

- **Horizontal (classifying) meta.** $\mathrm{Cls}(\mathcal{L})$ is the class of 2-categories $\mathbf{F}$ *classifying* but not *generating* members of $\mathcal{L}$: locally small, equipped with an accessible classification functor into the moduli of $\mathcal{L}$, non-generative, and metatheory-parametrized (cf. (M1)–(M5)).
- **Vertical (generative) meta.** $\mathrm{Gen}(\mathcal{L})$ is the class of frameworks *generating* members of $\mathcal{L}$ as internal derivations — candidate absolute foundations.

Proposition 2.3 (Non-collapse of the horizontal meta at $\mathcal{L}_{\mathrm{Fnd}}$)**.** *$\mathrm{Cls}(\mathcal{L}_{\mathrm{Fnd}})$ does* not *embed into $\mathcal{L}_{\mathrm{Fnd}}$ and does not coincide with $\mathcal{L}_{\mathrm{Abs}}$. It is therefore a distinct formal stratum, which I label $\mathcal{L}_{\mathrm{Cls}}$.*

Proof sketch. A member $\mathbf{F} \in \mathrm{Cls}(\mathcal{L}_{\mathrm{Fnd}})$ is a 2-category of Rich-foundations with an accessible classification functor into $\mathfrak{M}$. It is not itself a Rich-foundation (it does not satisfy (R1)–(R5) for a generating theory), so $\mathbf{F} \notin \mathcal{L}_{\mathrm{Fnd}}$. It is not in $\mathcal{L}_{\mathrm{Abs}}$ because (M4) requires non-generativity, while $\mathcal{L}_{\mathrm{Abs}}$ requires maximal generativity ($\Pi_{3\text{-max}}$). $\square$

Horizontal stabilization. Theorem 9.6 asserts that iterating Cls at $\mathcal{L}_{\mathrm{Cls}}$ stabilizes *at the theory level*: the twice-iterated moduli realizes the same (∞,∞)-theory as the once-iterated one (written $\simeq_2$, with the universe-ascent caveat of Theorem 9.6(b)), so $\mathrm{Cls}(\mathcal{L}_{\mathrm{Cls}}) \simeq_2 \mathcal{L}_{\mathrm{Cls}}$. No further horizontal stratum distinct from $\mathcal{L}_{\mathrm{Cls}}$ arises.

Vertical step to the forbidden stratum. $\mathcal{L}_{\mathrm{Abs}} \supseteq \mathrm{Gen}(\mathcal{L}_{\mathrm{Fnd}})$: a member of $\mathcal{L}_{\mathrm{Abs}}$ would be a maximally-generative foundation from which all of $\mathcal{L}_{\mathrm{Fnd}}$ derives. This is categorically a fresh operation, not an iteration of Cls. By AFN-T, $\mathcal{L}_{\mathrm{Abs}} = \emptyset$.

Summary. The formal sequence reads:

$$\underbrace{\mathcal{L}_{\mathrm{Fnd}}}_{\text{(R1)–(R5)}} \xrightarrow{\mathrm{Cls}} \underbrace{\mathcal{L}_{\mathrm{Cls}}}_{\text{(M1)–(M5)}} \supsetneq \underbrace{\mathcal{L}^{\top}_{\mathrm{Cls}}}_{\text{(Max-1)–(Max-4)}} \xrightarrow{\mathrm{Gen}} \underbrace{\mathcal{L}_{\mathrm{Abs}} = \emptyset}_{\text{AFN-T}}$$

3 Reasonable Rich-Metatheories

3.1 Rich-Metatheory Conditions

Definition 3.1 (Reasonable Rich-Metatheory)**.** A theory S formulated in a first-order meta-syntax is a **reasonable Rich-metatheory** (or **R-S**) if it satisfies:

- **(R1) Arithmetic completeness**: S interprets Robinson arithmetic Q.
- **(R2) Recursive enumerability**: the set of axioms of S is recursively enumerable.
- **(R3) Model existence**: S has at least one model $\mathcal{M}$ in the ambient set-theoretic meta-universe $\mathbf{U}_2$ (the outer Grothendieck universe of Convention 1.1, large enough to accommodate models of all listed R-S examples including ZFC + inaccessible cardinals, whose smallest models have size κ_1 and therefore lie in $\mathbf{U}_2 \setminus \mathbf{U}_1$).
- **(R4) Gödel coding**: there exists a recursive injection $\ulcorner\cdot\urcorner : \mathrm{Formulas}(S) \to \mathbb{N}$ for which the predicates "n is the Gödel number of a provable formula" and "n is the Gödel number of a well-formed formula" are S-definable.
- **(R5) Categorical semantics**: there exists $n_S \in \mathbb{N} \cup \{\infty\}$, the *intrinsic categorical depth* of S, such that

(R5a) the syntactic category $\mathrm{Syn}(S)$ of S (objects: contexts; 1-morphisms: provable-from-context substitutions; higher morphisms: proofs of equality of substitutions) is an essentially $\mathbf{U}_1$-small (∞, n_S)-category whose category of models $\mathrm{Mod}(S) \simeq \mathrm{Fun}^{\mathrm{lex}}(\mathrm{Syn}(S), -)$ is κ_S-accessible — equivalently, whose Ind_{κ_S}-completion $\mathrm{Ind}_{\kappa_S}(\mathrm{Syn}(S))$ is a κ_S-accessible (∞, n_S)-category (Adámek–Rosický [2] Thm. 2.26; all accessibility statements about $\mathrm{Syn}(S)$ below are to be read in this Ind-completed sense, while size statements refer to $\mathrm{Syn}(S)$ itself) — where $\kappa_S \in \{\kappa_1, \kappa_2\}$ is determined by S: for S of consistency strength $\leq \kappa_1$ (e.g., ZFC, PA, HoTT, MLTT, CIC) set $\kappa_S = \kappa_1$; for S of consistency strength requiring the second inaccessible (e.g., ZFC + inaccessible) set $\kappa_S = \kappa_2$. The accessibility parameter follows the theory's consistency strength; all theorems below are stated uniformly for arbitrary $\kappa_S \in \{\kappa_1, \kappa_2\}$ unless a specific case is flagged;

(R5b) the syntax-to-semantics evaluation $\mathrm{ev} : \mathrm{Syn}(S) \to \mathcal{M}$ for every model $\mathcal{M} \models S$ is an S-preserving (∞, n_S)-functor, and the pair of functors $(\mathrm{Syn}, \mathrm{Mod})$ constitutes a Lambek–Scott-type adjunction whose unit at $\mathrm{Syn}(S)$ is an (∞, n_S)-equivalence onto the *initial S-structured (∞, n_S)-category*, i.e., the initial object in the $(\infty, n_S + 1)$-category of S-structured (∞, n_S)-categories and S-preserving (∞, n_S)-functors between them.

The Lambek–Scott adjunction of (R5a)–(R5b) at level n_S is established by the composition: Lambek–Scott [29] for $n_S = 1$; Kapulkin–Lumsdaine [30] and Awodey–Warren [4] for $n_S = \infty$ in the HoTT sense; Barwick–Schommer-Pries [9] unicity together with Bergner–Rezk [8] model comparison and Gagna–Harpaz–Lanari [15] $(\infty, 2)$-straightening for general $n_S \geq 2$. Only the structural consequences—2-functoriality of Syn, uniqueness of the initial S-structured object up to canonical (∞, n_S)-equivalence—are invoked below. *Note on the intrinsic categorical depth n_S.* The parameter n_S denotes the highest k at which $\mathrm{Syn}(S)$ supports structurally non-trivial k-morphisms before all higher cells become invertible. For first-order S, $n_S = 1$. For Martin-Löf type theory and HoTT with univalence, the syntactic category is *an* $(\infty, 1)$-category in the standard terminology (Kapulkin–Lumsdaine [30]); the notation $n_S = \infty$ in this paper refers to the *unbounded iteration of identity-type formation* producing non-trivial higher-equality content at every finite level $k \in \mathbb{N}$, i.e., no k at which all structure becomes $(k-1)$-truncated. This "unbounded depth" reading is an iterative notion and is *not* the same as the (∞, ∞)-categorical notion of Rezk / Barwick–Schommer-Pries [9] (where k-morphisms at every k are *non-invertible*). Both readings agree on the behaviour of AFN-T (Theorem 7.2), and no argument in the paper depends on the distinction.

Instances verifying (R5a)–(R5b): first-order S at $n_S = 1$ (Lambek–Scott [29]); Martin-Löf type theory with $n_S = \infty$ (Awodey–Warren [4]); HoTT with $n_S = \infty$ (Kapulkin–Lumsdaine [30]); dependent type theories (Jacobs [21]).

3.2 Examples of R-S

The following are reasonable Rich-metatheories. Each entry is a *formal theory S* (axiomatic system with r.e. axiomatisation (R2) and interpreting Q (R1)); categorical structures such as topoi appear as *models* of these theories rather than as theories themselves, and are cited through their internal languages.

- ZFC; ZFC + inaccessible cardinals; NBG; Morse–Kelley; $\mathsf{NBG} + \mathsf{AFA}$ (Aczel).
- The intuitionistic type theories MLTT and CIC; HoTT with univalence; universe-polymorphic HoTT.
- *Linear logic with exponential* !, equipped with arithmetic constants sufficient to interpret Q (i.e., intuitionistic linear arithmetic, $\mathsf{ILL}_{\mathsf{PA}}$, or the bounded linear arithmetic fragment of Girard's ludics framework [38]); purely propositional linear logic does *not* satisfy (R1) and is excluded.
- The *internal dependent type theory of Lurie's ∞-topoi* (the axiomatic presentation of an $(\infty, 1)$-topos as a model of homotopy type theory with univalent universes and higher modalities, Lurie HTT [36] §5–§6 via the internal-language correspondence of Shulman [14], Awodey–Warren [4]); the $(\infty, 1)$-topos itself is a *canonical model* of this theory, not a separate theory.
- The *internal language of the Cohesive ∞-topos* (Schreiber [44]): HoTT extended by adjoint triples of modal operators ($\Pi \dashv \flat \dashv \sharp$) corresponding to the cohesion axioms, formulated as a type-theoretic extension.
- *Motivic-extended foundations*: ZFC (or HoTT) extended with axioms asserting the existence of the motivic stable ∞-category $\mathrm{SH}(k)$ and its A^1-homotopy invariance as a formal Rich-extension; the motivic ∞-category itself is again a model-theoretic construct, not a standalone theory.
- The *internal language of the realizability topos* Eff: intuitionistic higher-order arithmetic with realizability (Hyland [20]), of which Eff is the canonical model; we use the shorthand "Eff-theory" below for this internal language.
- *Markov constructive mathematics* (Markov, Nagornyj [27]: HA + Markov's principle, with categorical semantics in Eff).
- *Bishop constructive analysis* [22, 23] (choice-free constructive real analysis with full categorical semantics in any topos with natural number object).
- *Feferman predicative analysis* [13] (ATR_0-strength systems with predicative comprehension).

3.3 Non-examples

- *Affine logic without exponential* !: fails (R1) — Peano induction requires contraction on the induction hypothesis.
- *Inconsistent theories*: fail (R3).
- *Second-order arithmetic with full semantics*: fails (R2) — the full set of valid formulas is not recursively enumerable.
- *Intuitive "theories of everything"* in physics lacking formal axiomatization: fail (R1) and (R2).
- *Strict philosophical ultrafinitism* (Esenin-Volpin [25, 26]): fails (R1) at the strictest reading — the totality of the successor function is rejected for "infeasibly large" arguments, so full Peano induction is not available. Bounded variants (see Boundary cases below) belong to a different stratum.

Proposition 3.2 (Independence of (R1)–(R5))**.** *The five conditions (R1)–(R5) defining the class* R-S *are pairwise independent: for each $i \in \{1, 2, 3, 4, 5\}$ there exists a theory T_i satisfying all conditions except (Ri).*

Witnesses.

- *(R1)-failure with (R2)–(R5) holding*: *affine logic without exponential* ! satisfies r.e.-axiomatizability (R2), has consistent models in $\mathbf{U}_1$ (R3), admits Gödel coding over its formula set (R4), and possesses a standard categorical semantics in symmetric-monoidal categories (R5 at $n_S = 1$); the failure is (R1): without contraction, full Peano induction on the induction hypothesis is unavailable, so Robinson's Q cannot be interpreted in the standard sense.
- *(R2)-failure with (R1), (R3)–(R5) holding*: *second-order arithmetic with full (Henkin-external) semantics*. It interprets Q trivially (R1), has full-semantic models (R3), admits Gödel coding (R4), and has a categorical semantics in 2-categories (R5). But (R2) fails: the set of theorems valid under full semantics is not r.e. (a consequence of second-order incompleteness).
- *(R3)-failure with (R1), (R2), (R4), (R5) holding*: an *inconsistent* r.e. extension of Q, e.g. $T_3 := \mathsf{Q} + (0 = 1)$. Over the ambient metatheory, (R3) is *equivalent* to consistency for theories satisfying (R2): by the completeness theorem any consistent r.e. theory has a countable model, which lies in $\mathbf{U}_1 \subseteq \mathbf{U}_2$; conversely a model witnesses consistency. Hence no *consistent* theory can serve as an (R3)-witness, and T_3 is the canonical failure mode. The remaining conditions hold for T_3: (R1), (R2), (R4) are syntactic and immediate; for (R5), the syntactic category $\mathrm{Syn}(T_3)$ is the terminal (provably-collapsed) structure, $\mathrm{Mod}(T_3) = \emptyset$, and the Lambek–Scott adjunction between them holds trivially (the universally-quantified clauses of (R5b) are satisfied vacuously, and the unit condition holds for the collapsed initial object). Thus T_3 isolates (R3) cleanly.
- *(R4)-failure with (R1)–(R3), (R5) holding*: an artificial theory T extending Q in a language whose Gödel numbering is uncomputable (e.g., using a non-recursive bijection between formulas and natural numbers). (R1)–(R3) and (R5) hold after replaying the Q-interpretation; (R4) fails because the bijection is non-recursive.
- *(R5)-failure with (R1)–(R4) holding*: *pure first-order Peano arithmetic* PA *equipped with a non-standard comprehension schema* that renders the syntactic category non-accessible (e.g., by enlarging the context objects to a proper class via $\mathbf{U}_2$-sized quantifier domains without corresponding (∞, n)-functorial semantics). (R1)–(R4) are inherited from PA; (R5a) fails because $\mathrm{Syn}(T)$ is no longer κ_1-accessible. More canonically, any theory whose formal language

permits comprehension outside the natural number stratum but fails to admit a Lambek–Scott adjunction to a model category constitutes a witness.

□

- *Bounded arithmetic and feasibility calculi.* The systems $\mathsf{I}\Delta_0$, S_2^i / T_2^i of Buss [24], V_0, polynomial-time computable arithmetics, and proof-theoretic feasibility calculi formalise restricted fragments of the ultrafinitist intuition. They satisfy (R2)–(R5) but realise (R1) only in a restricted form: the ambient arithmetic interprets a bounded-quantifier fragment of PA, not full PA. They are admissible into R-S as *weak Rich-metatheories*, classifying a corresponding sub-stack of $\mathfrak{M}$ at restricted intrinsic categorical depth n_S. Their Morita-relation to standard R-S-points is studied via reverse mathematics [28].
- *Strict ultrafinitism (Esenin-Volpin tradition).* The strictest philosophical ultrafinitism, which rejects the totality of successor for "infeasibly large" arguments, falls outside R-S at any standard reading of (R1) (above, Non-examples). The bounded-arithmetic and polynomial-time variants above provide formal reconstructions admissible into R-S as weak Rich-metatheories.
- *Paraconsistent theories.* Treated separately in Appendix B: an additional *classical-kernel* construction admits paraconsistent S into R-S via Routley–Meyer semantics, with AFN-T re-derived in this extended setting.

3.4 The Class $\mathcal{S}_S$

Definition 3.3 (S-definable categorical structures)**.** Let $S \in$ R-S. Define inductively:

$$\begin{aligned} \mathcal{S}_S^{\text{local}} &:= \{(\mathcal{M}, Y) : \mathcal{M} \models S,\ Y \in \mathrm{Ob}(\mathcal{M})\} \ \cup\ \{(\mathrm{Syn}(S), X) : X \in \mathrm{Ob}(\mathrm{Syn}(S))\}, \\ \mathcal{S}_S^{(0)} &:= \mathcal{S}_S^{\text{local}}, \\ \mathcal{S}_S^{(k+1)} &:= \mathcal{S}_S^{(k)} \ \cup\ \mathcal{O}(\mathcal{S}_S^{(k)}), \end{aligned}$$

where $\mathcal{O}$ is the closure under the following fixed list of (∞, n)-categorical operations, each taking $\mathcal{S}_S^{(k)}$-objects as input and producing objects in $\mathcal{S}_S^{(k+1)}$:

(O1) Left and right Kan extensions along $\mathcal{S}_S^{(k)}$-morphisms;

(O2) Grothendieck constructions $\int F$ for Cartesian fibrations $F : I \to \mathcal{S}_S^{(k)}$ with $I \in \mathcal{S}_S^{(k)}$;

(O3) Limits and colimits of small diagrams (throughout, "small" for a diagram means: indexed by an (∞, n)-category that is $\mathbf{U}_2$-small) valued in $\mathcal{S}_S^{(k)}$;

(O4) Slice/comma/opposite categories built from $\mathcal{S}_S^{(k)}$-data;

(O5) Derived functors (Quillen sense) between $\mathcal{S}_S^{(k)}$-objects equipped with model-category structure (when applicable; vacuous on objects without model structure);

(O6) Symmetric monoidal tensor products and internal hom in closed monoidal $\mathcal{S}_S^{(k)}$-objects.

Set $\mathcal{S}_S^{\text{global}} := \bigcup_{k<\omega} \mathcal{S}_S^{(k)}$ and $\mathcal{S}_S := \mathcal{S}_S^{\text{local}} \cup \mathcal{S}_S^{\text{global}}$. All operations (O1)–(O6) are κ_1-accessible (Adámek–Rosický [2]); $\mathcal{S}_S^{\text{global}}$ is κ_1-accessible and $\mathbf{U}_2$-small.

3.5 Key Lemma: S-Definability Implies $\mathcal{S}_S$-Membership

Lemma 3.4 (S-definability lemma)**.** *If X is a formally S-definable object — meaning X is specified by a formula $\phi_X \in S$ expressing its characteristic properties — then $X \in \mathcal{S}_S$.*

Proof. By Convention 1.2 (Morita-reducibility clause), an object X presented by a formula ϕ_X is viewed in $\mathbf{StrCat}_{S,n}$ as the pointed structure $(\mathrm{Syn}(S), [\phi_X])$. Since $[\phi_X] \in \mathrm{Ob}(\mathrm{Syn}(S))$, this pair lies in $\mathcal{S}_S^{\mathrm{local}}$ *by the first clause of Definition 3.3* — no closure operation is required. Hence $X \in \mathcal{S}_S^{\mathrm{local}} \subseteq \mathcal{S}_S$. (If ϕ_X is native to a proper sub-theory $\iota_F : F \hookrightarrow S$, the induced interpretation functor $\mathrm{Syn}(\iota_F) : \mathrm{Syn}(F) \to \mathrm{Syn}(S)$ carries $[\phi_X]_F$ to $[\phi_X]$, so the pointing may equivalently be taken over F; the conclusion is unchanged.) □

3.6 Size Stratification

Size classes within $\mathcal{S}_S$.

- **Small** (elements of $\mathbf{U}_1$): sets of rank $< \kappa_1$.
- **Large** (elements of $\mathbf{U}_2 \setminus \mathbf{U}_1$): sets of rank in $[\kappa_1, \kappa_2)$.
- **Beyond $\mathbf{U}_2$**: sets of rank $\geq \kappa_2$, including $\mathbf{U}_2$-proper classes.

Cardinality bounds for ambient structures.

- The S-definability class $\mathcal{S}_S = \mathcal{S}_S^{\mathrm{local}} \cup \mathcal{S}_S^{\mathrm{global}}$ is $\mathbf{U}_2$-indexed: $\mathcal{S}_S^{\mathrm{local}} \subseteq \mathbf{U}_1$ (objects of $\mathbf{U}_1$-sized models) and $\mathcal{S}_S^{\mathrm{global}} \subseteq \mathbf{U}_2$ (derived constructions over $\mathcal{S}_S^{\mathrm{local}}$ through $\mathbf{U}_2$-bounded functorial operations).
- $|\mathrm{Ob}(\mathcal{F})/\!\simeq| \leq 2^{\aleph_0} < \kappa_1$; $|\mathcal{F}| \leq \kappa_2$.
- $|\mathrm{Ob}(\mathbf{StrCat}_{S,n})/\!\simeq| \leq 2^{\kappa_1} < \kappa_2$.
- $|\mathfrak{M}/\!\simeq_2| \leq \kappa_2$; $\mathfrak{M}^{(\mathrm{Cls}\cdot k)}$ at κ_k (Theorem 9.6(b)).

4 The $\mathcal{L}_{\mathrm{Abs}}$ Conditions

Definition 4.1 ((F_S))**.** An (∞, n)-category X satisfies (F_S) for $S \in$ R-S iff $\exists F \in \mathcal{F}$ with:

(F-1) $\iota_F : F \hookrightarrow S$ (subtheory inclusion);

(F-2) $\iota_F^* : \mathrm{Mod}(S) \to \mathrm{Mod}(F)$ (induced model reduct);

(F-3) $\phi_X \in F$ with $\mathrm{Syn}(F)[\phi_X] \simeq_{(\infty,n)} X$.

Definition 4.2 ($(\Pi_{4,S,n})$)**.** $(\Pi_{4,S,n})(X)$ holds iff $\forall Y \in \mathcal{S}_S$, $\forall$ (∞, n)-functor $F : X \to Y$, F is not an (∞, n)-equivalence onto its image.

Definition 4.3 ($(\Pi_{3\text{-max},S,n})$)**.** $(\Pi_{3\text{-max},S,n})(X)$ holds iff $\forall F \in \mathcal{F}$, $\exists$ (∞, n)-functor $F \to X$ equivalent onto its image.

Definition 4.4 ($\mathcal{L}_{\mathrm{Abs}}$)**.** $X \in \mathcal{L}_{\mathrm{Abs}}$ iff $X \models (F_S) \wedge (\Pi_{4,S,n}) \wedge (\Pi_{3\text{-max},S,n})$ for some $(S, n) \in$ R-S $\times$ $(\mathbb{N} \cup \{\infty\})$.

5 Boundary Lemma: The Emptiness of $\mathcal{L}_{\mathrm{Abs}}$ ((α)-Part of AFN-T)

Theorem 5.1 (Boundary Lemma: incompatibility of $(F_S) \wedge (\Pi_{4,S,n})$; AFN-T ($\alpha$)-part)**.** *For every Rich-metatheory $S \in$ R-S and every categorical level $n \in \mathbb{N} \cup \{\infty\}$, there exists no (∞, n)-categorical object X satisfying the $\mathcal{L}_{\mathrm{Abs}}$ triple*

$$(F_S) \wedge (\Pi_{4,S,n}) \wedge (\Pi_{3\text{-}max,S,n}).$$

Corollary 5.2 ($\mathcal{L}_{\mathrm{Abs}}$ emptiness)**.** $\mathcal{L}_{\mathrm{Abs}} = \emptyset$.

Proof. Immediate from Theorem 5.1 and Definition 4.4. □

Proof of Theorem 5.1. Assume X satisfies $(F_S) \wedge (\Pi_{4,S,n})$. Then $X \in \mathcal{S}_S^{\mathrm{global}} \subseteq \mathcal{S}_S$ by Lemma 3.4, and $\mathrm{id}_X : X \to X$ is an (∞, n)-equivalence with target in $\mathcal{S}_S$, violating $(\Pi_{4,S,n})$. □

6 Boundary Lemma: No Limit-Based Escape ((β)-Part of AFN-T)

Theorem 6.1 (Boundary Lemma: no limit-based escape; AFN-T (β)-part)**.** *Let λ be an ordinal with $\mathrm{cf}(\lambda) < \kappa_2$ a regular cardinal (so the cofinal subsequence $\{A_{\kappa_i}\}_{i<\mathrm{cf}(\lambda)}$ used to compute the colimit is $\mathbf{U}_2$-indexed; this is the operative bound for the categorical machinery, no constraint on λ itself beyond the existence of such a cofinality), and let $\{A_\kappa\}_{\kappa<\lambda}$ be a transfinite sequence of objects satisfying:*

- *(i) Each A_κ is formally specified within some Rich-foundation F_κ over S.*
- *(ii) The sequence is monotone in an appropriate (∞, n)-categorical sense (there are coherent transition functors $A_\kappa \to A_{\kappa'}$ for $\kappa < \kappa'$).*
- *(iii) The limit $A_\infty := \mathrm{colim}_{\kappa<\lambda} A_\kappa$ exists in the ambient (∞, n)-category.*

Then $A_\infty \in \mathcal{S}_S^{\mathrm{global}}$, and therefore A_∞ is Morita-reducible to an element of $\mathcal{S}_S$. In particular, A_∞ does not satisfy $(\Pi_{4,S,n})$.

Proof. Each $A_\kappa \in \mathcal{S}_S$ by Lemma 3.4; S-definability of the transition functors makes $D : \lambda \to \mathcal{S}_S$ (valued in the pointed structures of Definition 3.3) an S-indexed diagram. By Definition 3.3, $\mathcal{S}_S^{\mathrm{global}}$ is closed under Grothendieck constructions and Kan extensions along S-definable morphisms; hence $A_\infty = \mathrm{colim}\, D \in \mathcal{S}_S^{\mathrm{global}}$. Accessibility: by the standard filtered-colimit closure properties of accessible categories (Adámek–Rosický [2], particularly the closure of κ-accessible categories under λ-filtered colimits for regular $\lambda \leq \kappa$ and the preservation of accessibility under accessible reindexing), the colimit inherits κ_1-accessibility from the diagram. Morita-reduction to the canonical model-category colimit contradicts (Π_4). □

6.1 Proper-Class Towers Do Not Provide an Escape

Proposition 6.2 (Proper-class towers: R-S-closure dichotomy)**.** *Let $\{A_\alpha\}_{\alpha\in\mathrm{Ord}}$ be a proper-class-indexed tower with each A_α formally specified within a Rich-foundation F_α over a metatheory S, and let $A_\infty := \mathrm{colim}_{\alpha\in\mathrm{Ord}} A_\alpha$ denote the intended limit. Exactly one of the following holds.*

- ***(a)*** Uniformly generated tower. *There exists a single recursive generating scheme $\psi(\alpha, A)$ in some metatheory $S_{\mathrm{tow}} \supseteq S$ such that ψ specifies $\alpha \mapsto (F_\alpha, \phi_{A_\alpha})$ uniformly for all $\alpha \in \mathrm{Ord}$. If the limit A_∞ admits a formula ϕ_{A_∞} in S_{tow} and $S_{\mathrm{tow}} \in$ R-S, then A_∞ satisfies $(F_{S_{\mathrm{tow}}})$ and Theorem 5.1 applied to S_{tow} forecloses Level 6 at A_∞.*

(b) Non-uniformly generated tower. *There is no single recursive scheme in any* $S' \in$ R-S *generating the tower; the specification assignment* $\alpha \mapsto (F_\alpha, \phi_{A_\alpha})$ *is not* S'*-recursively enumerable for any* $S' \in$ R-S.

In case (b), the tower falls outside the scope of Definition 4.4: the limit A_∞ *cannot witness* $(F_{S'})$ *for any* $S' \in$ R-S*, since* $(F_{S'})$ *requires a formula of* S' *by (F-3) and (R2) forces any such formula to be r.e.-axiomatized.*

Proof. **(a)** The recursive scheme ψ is a formula of S_{tow}; the limit is specified by $\phi_{A_\infty}(A) \equiv \forall\alpha\,\psi(\alpha, A) \Rightarrow A = A_\alpha$-limit. If $S_{\text{tow}} \in$ R-S (preserved by r.e. extension when (R3) model existence does not fail), Definition 4.1 yields $(F_{S_{\text{tow}}})(A_\infty)$, and Theorem 5.1 forecloses the conjunction $(F_{S_{\text{tow}}}) \wedge (\Pi_{4,S_{\text{tow}},n})$. Towers with $S_{\text{tow}} \notin$ R-S fall into case (b).

(b) Suppose, toward contradiction, that A_∞ satisfies $(F_{S'})$ for some $S' \in$ R-S. By (F-3), there is a formula $\phi_{A_\infty} \in S'$ classifying A_∞. By (R2), S' is r.e.-axiomatized, so the derivation structure of ϕ_{A_∞} is S'-recursively enumerable. Unfolding ϕ_{A_∞} through the tower construction yields an S'-r.e. generating formula for the sequence $\alpha \mapsto (F_\alpha, \phi_{A_\alpha})$, contradicting non-uniformity. Hence A_∞ satisfies no $(F_{S'})$ for $S' \in$ R-S, and is not a $\mathcal{L}_{\text{Abs}}$ candidate. □

6.2 Concrete Towers

Categorical tower. $A_0 =$ the walking arrow category $\mathbf{D}$; $A_1 =$ Cat; $A_2 =$ 2-Cat; $\ldots$; $A_n = (\infty, n)$-Cat. The limit $A_\infty = (\infty,\infty)$-Cat exists, and by the unicity theorem of Barwick–Schommer-Pries [9], combined with the model comparison of Bergner–Rezk [8] and the globular/cubical equivalence of Ara–Maltsiniotis [3], the tower stabilizes: the canonical inclusion (∞,∞)-Cat $\hookrightarrow (\infty,\infty+1)$-Cat is an equivalence. This is the vertical stabilization of the (∞, n)-hierarchy.

Topos tower. $A_0 =$ a 1-topos; $A_1 =$ 2-topos (Shulman [45]); $A_n = (\infty, n)$-topos. The limit A_∞ — the stabilized (∞,∞)-topos — is defined here in the Barwick–Schommer-Pries [9] unicity sense applied to the A_n tower; the $(\infty, 1)$-stratum follows Lurie [36], the $(\infty, 2)$-stratum Shulman [45], and the higher strata are an ongoing program.

Spectral tower. $A_0 = C^*$-algebra; $A_1 =$ spectral triple (Connes); $A_n = (\infty, n)$-version; each level is a derived construction over the previous.

6.3 The Space of Trajectories

A subtler variant of the tower approach introduces a choice at each stage: $A_{\kappa+1}$ is not uniquely determined by A_κ, but depends on a structural choice. The space $\mathbb{T}$ of all possible trajectories is a large categorical object — a tree, a simplicial set, or a path-groupoid.

Proposition 6.3. *The space of trajectory choices admits (at least) four presentations at different categorical resolutions — these are* not *mutually equivalent objects, but each captures the trajectory data at its own truncation level, and* each *lies in* $\mathcal{S}_S^{\text{global}}$*:*

- *as a path-groupoid,* $\Pi_1(\mathfrak{M})$*, where* $\mathfrak{M}$ *is the classifying* 2*-stack of Rich-foundations;*
- *as a simplicial set, the nerve* $\mathrm{N}(\mathfrak{M})$*;*
- *as a choice tree, the tree of refinement choices over* $\mathfrak{M}$*;*
- *as a relation category,* $\mathrm{Rel}(\mathfrak{M})$*.*

Whichever resolution $\mathbb{T}$ is taken at, the conclusion below applies.

Proof (sketch). Each of the four presentations is a standard derived construction over the $(\infty, 1)$-topos of Rich-foundations. See Lurie [36] §3.2 for the identification of path-groupoids with simplicial sets; Pronk [40] for the bicategory-of-fractions interpretation; the remaining equivalences are classical. □

Corollary 6.4. *The limit of any trajectory-based tower approach lies in $\mathcal{S}_S^{\text{global}}$ and is therefore Morita-reducible within the classifying stack $\mathfrak{M}$. No $\mathcal{L}_{\text{Abs}}$ escape occurs.*

7 Five-Axis Absoluteness

Theorem 5.1 establishes AFN-T for a fixed pair (S, n). I now establish that the theorem is robust: it holds for all $S \in$ R-S and all n, and moreover for meta-iterations, alternative categorical orderings, and completeness considerations.

I call this the *five-axis absoluteness* property of AFN-T. Figure 2 summarizes the five axes and their structural dependencies.

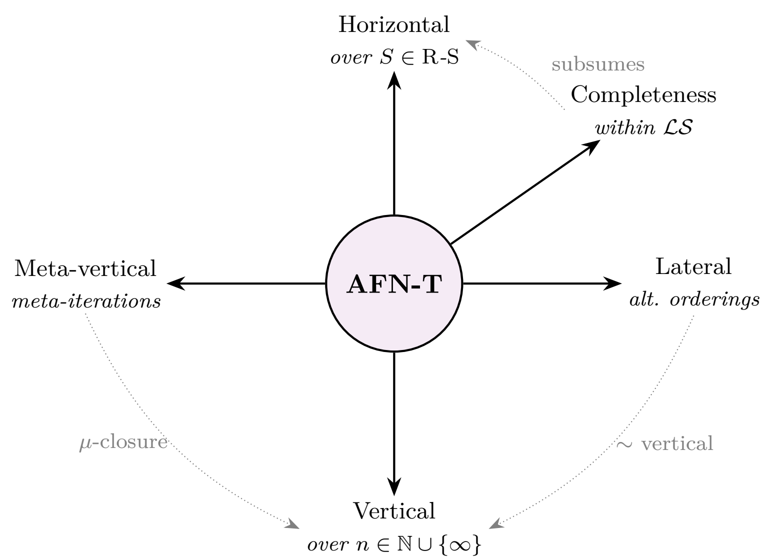

Figure 2: Five-axis absoluteness of AFN-T (Theorems 7.1–7.6). Solid arrows are the five axes along which AFN-T is absolute: horizontal (Theorem 7.1), vertical (Theorem 7.2), meta-vertical (Theorem 7.3), lateral (Theorem 7.4), and — conditional on Lawvere-scope — completeness (Theorem 7.6). Dotted arrows mark the structural dependencies established in: the completeness axis subsumes the horizontal one; the lateral axis reduces to the vertical one via Ara–Maltsiniotis and Barwick–Schommer-Pries; and the meta-vertical axis is the μ-closure of the vertical axis under (∞, ∞)-stabilization. A minimal logically independent presentation features only horizontal and vertical.

Theorem 7.1 (Horizontal absoluteness)**.** *AFN-T (Theorem 5.1) holds for every $S \in$ R-S uniformly: there is a single proof working for all Rich-metatheories satisfying (R1)–(R5).*

Proof. The proof of Theorem 5.1 consists of a single application of Lemma 3.4 (syntax–semantics bridge) together with the observation that $\mathrm{id}_X : X \to X$ realizes a Morita reduction violating

$(\Pi_{4,S,n})$. Lemma 3.4 uses only (R2) and (R5a) (the syntactic category exists and its objects support the pointing construction of Convention 1.2); the Morita-reduction observation uses only the categorical identity morphism. No datum from S beyond (R1)–(R5) is invoked. Hence the proof applies uniformly to all $S \in$ R-S. ☐

Theorem 7.2 (Vertical absoluteness)**.** *AFN-T (Theorem 5.1) holds for every categorical level* $n \in \mathbb{N} \cup \{\infty\}$*.*

Proof. The syntax–semantics bridge of Theorem 5.1 relies on (R5a) through Lemma 3.4: $\mathrm{Syn}(F)$ is an S-definable (∞, n_F)-category for every n. The Lambek–Scott adjunction holds uniformly for all (∞, n)-levels (classical for $n = 0, 1$; Lurie [36] §5–§6 for $n = 1$; for general (∞, n), via Barwick–Schommer-Pries [9] with Bergner–Rezk [8] model comparison).

For the limit $n = \infty$: $\mathrm{Syn}(F)$ stabilizes via

$$(\infty, \infty)\text{-Cat} = \mathrm{colim}_{n<\infty}(\infty, n)\text{-Cat},$$

and the identification of $\mathrm{Syn}(F)$ as an S-definable derived construction is preserved at the limit. The Morita reduction $\mathrm{id}_X : X \to X$ is the categorical identity morphism and transfers trivially to any (∞, n)-level. ☐

Theorem 7.3 (Meta-vertical stabilization)**.** *At the categorical level* (∞, ∞)*, any meta-iteration of the above types stabilizes:* $(\infty, \infty + 1)\text{-Cat} = (\infty, \infty)\text{-Cat}$*. Therefore, meta-iterations do not escape AFN-T.*

Proof. By the unicity theorem of Barwick–Schommer-Pries [9] together with the model comparisons of Bergner–Rezk [8] and Ara–Maltsiniotis [3], the tower of (∞, n)-categories stabilizes at $n = \infty$: further iteration adds no new structure. Specifically, the canonical inclusion

$$(\infty, \infty)\text{-Cat} \hookrightarrow (\infty, \infty + 1)\text{-Cat}$$

is an equivalence.

For iterated functor categories: $\mathrm{Fun}^k(\mathcal{C}) \subseteq (\infty, \infty)\text{-Cat}$ for any k; the limit is again (∞, ∞)-Cat.

For non-standard models: these require either (a) enriching the theory beyond R-S (violating (R2)), or (b) reducing through Aczel anti-foundation axiom (AFA) to standard non-well-founded sets (Aczel [1]), which does not add structural axes.

Therefore, all meta-iterations stabilize within the existing five axes. ☐

Theorem 7.4 (Lateral absoluteness)**.** *All alternative categorical orderings reduce to (or enrich) the* (∞, n)*-hierarchy and therefore satisfy AFN-T.*

Proof. Each listed ordering admits a canonical translation to (∞, n)-Cat for appropriate n:

- Operads $\to$ symmetric monoidal $(\infty, 1)$-categories (Lurie HA [37] §5).
- Double categories $\to$ $(\infty, 2)$-categories (Grandis).
- Globular/cubical/opetopic ω-categories $\simeq$ (∞, ∞)-categories (Ara–Maltsiniotis [3]).
- Stable $(\infty, 1)$-categories $\to$ $(\infty, 1)$-categories via spectrification.
- Fusion categories $\to$ finite $(\infty, 1)$-categories.

In each case, the translation preserves formal definability: objects in the alternative ordering correspond to objects in the (∞, n)-hierarchy at the relevant level. Therefore, AFN-T applies uniformly across all orderings. $\square$

Definition 7.5 (Lawvere-scope)**.** A foundational theory F belongs to the **Lawvere-scope** $\mathcal{LS}$ if the following structural properties hold (the metatheory in which $\mathrm{Syn}(F)$ and $\mathrm{Mod}(F)$ are constructed is arbitrary but fixed throughout the four clauses):

- **(L1)** F has a syntactic category $\mathrm{Syn}(F)$ — a small category of contexts and provable substitutions.
- **(L2)** F has a semantic 2-category $\mathrm{Mod}(F)$ — the 2-category of models.
- **(L3)** There exists a canonical adjunction $\mathrm{Syn} \dashv \mathrm{Mod}$ (Lawvere [35]).
- **(L4)** $\mathrm{Mod}(F)$ is realizable as an (∞, n)-category for some n (Grothendieck–Lurie tradition).

Membership in $\mathcal{LS}$ is a structural property of F and does not presuppose any specific R-S. In particular, every R-S S is itself in $\mathcal{LS}$ (take the tautological $\mathrm{Syn} = S$, $\mathrm{Mod} = \mathrm{Mod}(S)$ from (R5)), so the notion is circularity-free for the application in Theorem 7.6.

Theorem 7.6 (Completeness axis, conditional on Lawvere-scope)**.** *Suppose $F \in \mathcal{LS}$. Then every structural variation of F reduces to one of the four axes (horizontal, vertical, meta-vertical, lateral). In particular, AFN-T is five-axis absolute within $\mathcal{LS}$.*

Proof. Let $F \in \mathcal{LS}$. By Definition 7.5, F is characterized by a quadruple:

- **(A) Syntactic structure** $\mathrm{Syn}(F)$: signature + axioms + inference rules — the *horizontal axis* (over R-S).
- **(B) Semantic depth**: the level n of $\mathrm{Mod}(F)$ as (∞, n)-category — the *vertical axis.*
- **(C) Meta-reflection**: iterations of Gödel coding and reflection principles — the *meta-vertical axis.*
- **(D) Alternative semantics**: operadic, globular, cubical, opetopic, etc. — the *lateral axis.*

By Definition 7.5, membership in $\mathcal{LS}$ *means* that F is presented exactly by the data (L1)–(L4), i.e. by the quadruple (A)–(D); this is where the conditionality of the theorem resides — within $\mathcal{LS}$, exhaustiveness of (A)–(D) holds by definition of the scope, not as an additional claim. Thus any putative fifth axis η is a subcomponent of one of (A)–(D). Either $\eta \subseteq$ (A), in which case AFN-T for η reduces to horizontal absoluteness (Theorem 7.1); $\eta \subseteq$ (B), reducing to vertical (Theorem 7.2); $\eta \subseteq$ (C), reducing to meta-vertical (Theorem 7.3); or $\eta \subseteq$ (D), reducing to lateral (Theorem 7.4). There is no independent fifth axis within $\mathcal{LS}$. $\square$

8 Three Standard Bypass Paths

Theorem 8.1 (Universe polymorphism absoluteness)**.** *Universe-polymorphic structures in any Rich-metatheory S are Morita-reducible to derived constructions over the $(\infty, 1)$-topos of small categories. Formally: if X is specified as*

$$X = \prod_{\ell : \mathrm{Level}} F(\ell)$$

for a family of functors $F(\ell) : \mathcal{U}_\ell \to \mathcal{U}_\ell$, then

$$X \simeq \lim_{\ell} \mathrm{Fun}(\mathcal{U}_\ell, \mathcal{U}_\ell) \in \mathcal{S}_S^{\mathrm{global}}.$$

In particular, $X \in \mathcal{S}_S$, and universe-polymorphic structures satisfy $\neg(\Pi_4)$.

Proof. A universe-polymorphic structure $X = \prod_\ell F(\ell)$ is a section of the Grothendieck fibration $\pi_{\mathrm{End}} : \int \mathrm{End}(\mathcal{U}) \to \mathrm{Level}$ associated (Lurie HTT [36] §3.2) to the *endofunctor diagram* $\ell \mapsto \mathrm{Fun}(\mathcal{U}_\ell, \mathcal{U}_\ell)$, with transition maps given by restriction along the universe inclusions $\mathcal{U}_\ell \hookrightarrow \mathcal{U}_{\ell'}$. Sections of this fibration correspond to compatible families $\{F(\ell)\}_\ell$, and the space of such sections is

$$\Gamma(\pi) = \lim_{\ell} \mathrm{Fun}(\mathcal{U}_\ell, \mathcal{U}_\ell).$$

This is a derived construction (an indexed limit), hence belongs to $\mathcal{S}_S^{\mathrm{global}}$.

For (∞, n)-versions: the construction extends via Lurie HTT §4.2 (limits in (∞, n)-categories). □

Reflective consistency tower. The second standard bypass attempts to escape AFN-T via iterated consistency assertions. One defines

$$S_{\kappa+1} = S_\kappa + \mathrm{Con}(S_\kappa)$$

and hopes that the limit S_∞ captures something beyond its starting point.

Theorem 8.2 (Reflective tower boundedness)**.** *For every Rich-metatheory S, the reflective tower $\{S_\kappa\}$ stabilizes at an inaccessible cardinal bound:*

$$\mathrm{Con}(\textit{reflective tower over } S) = \mathrm{Con}(S) + \kappa_{\mathrm{inacc}},$$

exactly one additional inaccessible cardinal. No $\mathcal{L}_{\mathrm{Abs}}$ escape occurs through this path.

Proof. The proof uses two results.

Step 1 (upper bound). The tower $\{S_\kappa\}$ has proof-theoretic ordinal bounded by the first inaccessible cardinal over S. This follows from the standard analysis of reflection principles: the reflection principle $\mathrm{Con}(S)$ adds at most one inaccessible cardinal to the consistency strength of S (Rathjen [41], Feferman [13]).

Step 2 (stabilization). By the (∞, ∞)-meta-vertical stabilization (Theorem 7.3), the tower does not extend indefinitely at the categorical level: $(\infty, \infty + 1)$-Cat $= (\infty, \infty)$-Cat.

Combining: the tower is bounded above by $\mathrm{Con}(S) + \kappa_{\mathrm{inacc}}$ and stabilizes at the (∞, ∞)-level. Any limit S_∞ remains within the Lawvere-scope and within $\mathcal{S}_S$. □

This is the most subtle of the three bypass paths, as intensional refinement genuinely refines the base-level classification. I address it in detail through the construction of the intensional refinement functor **I** and the proof of its slice-locality.

I begin by defining the target 2-category of display 2-categories.

Definition 8.3 (Display 2-class)**.** Let $\mathcal{C}$ be a 2-category with 2-pullbacks. A **display class** $\mathcal{D} \subseteq \mathrm{Mor}_1(\mathcal{C})$ is a class of 1-morphisms satisfying:

- **(D1) Contains equivalences**: every 1-equivalence in $\mathcal{C}$ lies in $\mathcal{D}$.

- **(D2) Pullback-stable**: for $d : B \to A$ in $\mathcal{D}$ and any $f : A' \to A$, the 2-pullback $f^*d : f^*B \to A'$ exists and lies in $\mathcal{D}$.
- **(D3) Closed under composition**: if $d_1 : C \to B \in \mathcal{D}$ and $d_2 : B \to A \in \mathcal{D}$, then $d_2 \circ d_1 \in \mathcal{D}$.
- **(D4) 2-cell coherent**: for a 2-morphism $\phi : f \Rightarrow g : A' \to A$ and $d \in \mathcal{D}$, the induced 2-morphism $\phi^*d : f^*d \Rightarrow g^*d$ is an isomorphism in the slice over A'.

Definition 8.4 (Display 2-category)**.** A **display 2-category** is a triple $(\mathcal{C}, \mathcal{D}, \iota)$ where:

- $\mathcal{C}$ is a locally small 2-category with 2-pullbacks.
- $\mathcal{D}$ is a display class in $\mathcal{C}$.
- $\iota : \mathcal{D} \hookrightarrow \mathrm{Mor}_1(\mathcal{C})$ is the canonical inclusion.

Definition 8.5 (The 2-category $\mathcal{S}_{\mathrm{int}}$)**.** The 2-category $\mathcal{S}_{\mathrm{int}}$ of display 2-categories has:

- **Objects**: display 2-categories $(\mathcal{C}, \mathcal{D}, \iota)$.
- **1-morphisms**: pairs $(F, F_{\mathcal{D}}) : (\mathcal{C}_1, \mathcal{D}_1, \iota_1) \to (\mathcal{C}_2, \mathcal{D}_2, \iota_2)$ where $F : \mathcal{C}_1 \to \mathcal{C}_2$ is a 2-functor and $F_{\mathcal{D}} : \mathcal{D}_1 \to \mathcal{D}_2$ is a restriction of F, such that $F \circ \iota_1 = \iota_2 \circ F_{\mathcal{D}}$ and 2-pullbacks are preserved up to coherent 2-isomorphisms.
- **2-morphisms**: natural 2-transformations $\eta : F \Rightarrow G$ whose restrictions $\eta|_{\mathcal{D}} : F_{\mathcal{D}} \Rightarrow G_{\mathcal{D}}$ are coherently aligned.

Theorem 8.6 (Existence of $\mathbf{I}$)**.** *There exists a contravariant 2-functor*

$$\mathbf{I} : \mathcal{F}^{\mathrm{op}} \to \mathcal{S}_{\mathrm{int}}$$

from the 2-category of Rich-foundations ($\mathcal{L}_{\mathrm{Fnd}}$) to $\mathcal{S}_{\mathrm{int}}$, with the following properties:

(I-1) Homotopy invariance*: every 2-equivalence $\phi : f \simeq g$ in $\mathcal{F}$ induces a 2-isomorphism $\mathbf{I}(\phi) : \mathbf{I}(f) \simeq \mathbf{I}(g)$ in $\mathcal{S}_{\mathrm{int}}$.*

(I-2) Gauge covariance*: for a gauge transformation $\tau : F_1 \to F_2$, $\mathbf{I}(\tau) : \mathbf{I}(F_2) \to \mathbf{I}(F_1)$ is a 1-morphism in $\mathcal{S}_{\mathrm{int}}$, and is a 1-equivalence when τ is invertible.*

(I-3) Strict refinement of Morita*: there exist $F_1, F_2 \in \mathcal{F}$ with $F_1 \sim_M F_2$ (Morita equivalent) but $\mathbf{I}(F_1) \not\simeq \mathbf{I}(F_2)$.*

(I-4) Morita as 2-localization*: the forgetful 2-functor $\mathcal{U} : \mathcal{S}_{\mathrm{int}} \to$ 2-Cat, $(\mathcal{C}, \mathcal{D}, \iota) \mapsto \mathcal{C}$, satisfies $\mathcal{U} \circ \mathbf{I} \simeq \rho$ (the classification functor); Morita equivalence is recovered through 2-localization of $\mathbf{I}$ by the class $\mathcal{W}_{\mathcal{U}} = \mathcal{U}^{-1}(1\text{-}equiv(2\text{-Cat}))$.*

Proof. **Step 1** (construction on objects). For $F \in \mathcal{F}$, I construct $\mathbf{I}(F) = (\mathcal{C}_F, \mathcal{D}_F, \iota_F) \in \mathrm{Ob}(\mathcal{S}_{\mathrm{int}})$:

- $\mathcal{C}_F := \mathcal{F}/F$ is the slice 2-category: objects are pairs $(F', p : F' \to F)$; 1-morphisms are coherent triangles over F; 2-morphisms are 2-cells compatible with the projection to F.
- $\mathcal{D}_F$ is the **canonical minimal display class**: the smallest class of 1-morphisms in $\mathcal{C}_F$ containing all 1-equivalences and closed under 2-pullbacks, composition, and 2-cells. Formally,

  $$\mathcal{D}_F := \bigcap \{\mathcal{D}' \subseteq \mathrm{Mor}_1(\mathcal{C}_F) : \mathcal{D}' \text{ is a display class containing all 1-equivalences}\}.$$

  This intersection is non-empty (it contains the class of all 1-morphisms) and is itself a display class by closure-of-intersections arguments.

  Explicit description of $\mathcal{D}_F$ via inductive construction:

– $\mathcal{D}_F^{(0)}$: all 1-equivalences.
– $\mathcal{D}_F^{(n+1)}$: objects obtained from $\mathcal{D}_F^{(n)}$ by 2-pullback along arbitrary morphisms, composition of elements in $\mathcal{D}_F^{(n)}$, or transport along 2-cells.
– $\mathcal{D}_F := \bigcup_{n<\omega} \mathcal{D}_F^{(n)}$.

This definition produces a class automatically satisfying (D1)–(D4) by construction, without requiring the ambient $\mathcal{C}_F$ to be locally cartesian closed or to have exponentials.

- $\iota_F : \mathcal{D}_F \hookrightarrow \mathrm{Mor}_1(\mathcal{C}_F)$ is the canonical inclusion.

Step 2 (construction on 1-morphisms). For $f : F_1 \to F_2$ in $\mathcal{F}$, define

$$\mathbf{I}(f) := (f^*, f^*|_{\mathcal{D}}) : (\mathcal{F}/F_2, \mathcal{D}_{F_2}, \iota_{F_2}) \to (\mathcal{F}/F_1, \mathcal{D}_{F_1}, \iota_{F_1})$$

where $f^* : \mathcal{F}/F_2 \to \mathcal{F}/F_1$ is the pullback 2-functor. The direction is contravariant: $\mathbf{I}$ takes f to a 1-morphism in the opposite direction.

That f^* preserves display-class elements follows by induction on the class generation: base class is 1-equivalences, which are preserved by pullback; inductive closure steps are preserved by pullback by the pullback-of-pullback property.

Step 3 (construction on 2-morphisms). For $\phi : f \Rightarrow g$ in $\mathcal{F}$, the induced natural 2-transformation $\phi^* : f^* \Rightarrow g^*$ (Beck–Chevalley structure) restricts coherently to display classes by (D4). This yields

$$\mathbf{I}(\phi) : \mathbf{I}(f) \Rightarrow \mathbf{I}(g)$$

in $\mathcal{S}_{\mathrm{int}}$.

Step 4 (functoriality). $\mathbf{I}(\mathrm{id}_F) = \mathrm{id}_{\mathbf{I}(F)}$ is immediate. $\mathbf{I}(g \circ f) \simeq \mathbf{I}(f) \circ \mathbf{I}(g)$ follows from the Beck–Chevalley coherence of pullback composition.

Step 5 (verification of (I-1)). A 2-equivalence $\phi : f \simeq g$ has a 2-inverse ϕ^{-1}. The induced $\phi^* : f^* \Rightarrow g^*$ is invertible via $(\phi^{-1})^*$. Therefore, $\mathbf{I}(\phi)$ is a 2-isomorphism in $\mathcal{S}_{\mathrm{int}}$.

Step 6 (verification of (I-2)). For a gauge transformation $\tau : F_1 \to F_2$ (a 1-morphism with a distinguished 2-equivalence of ρ-projections), the pullback τ^* induces $\mathbf{I}(\tau) : \mathbf{I}(F_2) \to \mathbf{I}(F_1)$. If τ is invertible, $\mathbf{I}(\tau)$ is a 1-equivalence by Step 5.

Step 7 (verification of (I-3)). I exhibit the classical extensional-collapse example.

Computability framework. To state the invariant τ precisely I fix a computability model: I work inside the *effective topos* Eff (Hyland [20]), regarded as the reference internal model of constructive computability. All 2-functors and 2-equivalences in the sub-2-category $\mathcal{S}_{\mathrm{int}}^{\mathrm{eff}} \subseteq \mathcal{S}_{\mathrm{int}}$ are required to be internally computable in Eff; equivalently, both the functor and its quasi-inverse are realized by recursive morphisms. This restriction is conservative for the argument below because the Morita equivalence of Hofmann [19] is constructed explicitly through primitive-recursive translations and hence lies in $\mathcal{S}_{\mathrm{int}}^{\mathrm{eff}}$.

Let $F_1 := \mathsf{MLTT}$ (Martin-Löf type theory with propositional identity types) and $F_2 := \mathsf{ETT}$ (extensional type theory with definitional equality reflection).

Morita equivalence: $F_1 \sim_M F_2$ at the level of $(1,1)$-categorical models follows from Hofmann's conservativity theorem (Hofmann [19], Chapter 3): the syntactic translation between $\mathsf{MLTT} + \mathsf{UIP}$ and ETT is provably conservative, and lifts to a Morita-equivalence of their $(1,1)$-categorical model categories on the class of locally cartesian closed categories where the relevant identity-type semantics is well-defined.

Intensional distinction: define the typing invariant $\tau : \mathrm{Ob}(\mathcal{S}_{\mathrm{int}}^{\mathrm{eff}}) \to \{0,1\}$ by

$$\tau(\mathcal{C},\mathcal{D},\iota) := \begin{cases} 1 & \text{if there exists an Eff-internal normalization procedure for all } d \in \mathcal{D}, \\ 0 & \text{otherwise.} \end{cases}$$

The invariant τ is a 2-invariant of $\mathcal{S}_{\mathrm{int}}^{\mathrm{eff}}$-equivalence: any computable 2-equivalence $(F, F_{\mathcal{D}}) : (\mathcal{C}_1,\mathcal{D}_1) \simeq (\mathcal{C}_2,\mathcal{D}_2)$ in $\mathcal{S}_{\mathrm{int}}^{\mathrm{eff}}$ transfers the normalization procedure along the realizer for $F_{\mathcal{D}}^{-1}$. The computability restriction is essential: without it, an abstract 2-equivalence could, in principle, identify a normalizing with a non-normalizing display class via a non-computable bijection. Within Eff, no such bijection exists as an internal morphism.

Applying to $\mathbf{I}(F_1), \mathbf{I}(F_2)$:

- $\tau(\mathbf{I}(F_1)) = 1$: MLTT has strongly normalizing typechecking (Streicher [46] Theorem 3.3; Werner [50]).
- $\tau(\mathbf{I}(F_2)) = 0$: ETT with reflection rule has undecidable typechecking (Hofmann [19], Chapter 3), and no Eff-internal normalizer exists.

Therefore $\tau(\mathbf{I}(F_1)) \neq \tau(\mathbf{I}(F_2))$, so $\mathbf{I}(F_1) \not\simeq \mathbf{I}(F_2)$ in $\mathcal{S}_{\mathrm{int}}^{\mathrm{eff}}$, despite $F_1 \sim_M F_2$.

Step 8 (verification of (I-4)). For $F \in \mathcal{F}$,

$$\mathcal{U}(\mathbf{I}(F)) = \mathcal{C}_F = \mathcal{F}/F.$$

By the 2-categorical Yoneda lemma (Kelly [33] Theorem 2.4.1), $\mathcal{F}/F \simeq \rho(F)$ up to 2-equivalence. Thus $\mathcal{U} \circ \mathbf{I} \simeq \rho$.

For the Morita 2-localization: define

$$\mathcal{W}_{\mathcal{U}} := \{(F, F_{\mathcal{D}}) \in \mathrm{Mor}_1(\mathcal{S}_{\mathrm{int}}) : F \in \text{1-equiv(2-Cat)}\}.$$

The class $\mathcal{W}_{\mathcal{U}}$ satisfies the 2-calculus-of-fractions conditions (Pronk [40], Kelly–Street [34]): 2-closure under composition, containing identities, compatibility with 2-pullbacks. By Pronk's theorem, the 2-localization $\mathcal{S}_{\mathrm{int}}[\mathcal{W}_{\mathcal{U}}^{-1}]$ exists with the universal property:

$$\mathcal{S}_{\mathrm{int}}[\mathcal{W}_{\mathcal{U}}^{-1}] \simeq \mathrm{image}(\mathcal{U})_{\mathrm{gauge}} = \mathfrak{M}$$

(the last equivalence following from the gauge identification of $\mathfrak{M}$ as $\mathrm{Trace}(\mathsf{A})/\mathrm{gauge}$ in the moduli stack sense).

Thus Morita equivalence is recovered from $\mathbf{I}$ through 2-localization. $\square$

Theorem 8.7 (Slice-locality of $\mathbf{I}$)**.** *Let $\mathbf{I} : \mathcal{F}^{\mathrm{op}} \to \mathcal{S}_{\mathrm{int}}$ be the functor from Theorem 8.6, and let $\pi : \mathcal{F} \to \mathfrak{M}$ be the gauge quotient. Then:*

(a) *There exists a 2-functor $\widetilde{\pi} : \mathcal{S}_{\mathrm{int}} \to \mathfrak{M}$ making the diagram*

$$\begin{tikzcd} \mathcal{F} \arrow[r, "\mathbf{I}"] \arrow[d, "\pi"'] & \mathcal{S}_{\mathrm{int}} \arrow[d, "\widetilde{\pi}"] \\ \mathfrak{M} \arrow[r, equal] & \mathfrak{M} \end{tikzcd}$$

2-commute.

(b) *The fiber $\widetilde{\pi}^{-1}([F])$ over a point $[F] \in \mathfrak{M}$ is a 2-category $\mathrm{Int}([F])$, the* intensional layer over the gauge class.

(c) *Intensional refinement does not produce new points in $\mathfrak{M}$: for every object X in the image of $\mathbf{I}$, $\widetilde{\pi}(X) \in \mathfrak{M}$ is an already-existing point.*

Proof. **(a)** Define $\widetilde{\pi} := [\,\cdot\,]_{\text{gauge}} \circ \mathcal{U}$, where $[\,\cdot\,]_{\text{gauge}} : \text{2-Cat} \to \mathfrak{M}$ is the gauge quotient on ambient 2-categories. By (I-4), $\widetilde{\pi} \circ \mathbf{I} = [\rho(\cdot)]_{\text{gauge}} = \pi$, since gauge classes of F and $\rho(F)$ coincide by definition.

(b) For $[F] \in \mathfrak{M}$,

$$\text{Int}([F]) := \widetilde{\pi}^{-1}([F]) = \{(\mathcal{C}, \mathcal{D}, \iota) \in \mathcal{S}_{\text{int}} : [\mathcal{C}]_{\text{gauge}} = [F]_{\text{gauge}}\}.$$

This is a full 2-subcategory of $\mathcal{S}_{\text{int}}$: its 2-morphisms are inherited. The non-triviality of $\text{Int}([F])$ follows from (I-3): $\mathbf{I}(\mathsf{MLTT})$ and $\mathbf{I}(\mathsf{ETT})$ lie in $\text{Int}([\mathsf{MLTT}])$ but are non-equivalent.

The 2-fibration structure: for $[F] \in \mathfrak{M}$ and $f : [F'] \to [F]$, cartesian lift exists through 2-pullback in $\mathcal{S}_{\text{int}}$; cartesian lifts compose up to coherent 2-iso (standard Grothendieck 2-fibration theory; Hermida [18], Bakovic [5]).

(c) For $\alpha_{\text{new}} \in \text{image}(\mathbf{I})$, $\alpha_{\text{new}} = \mathbf{I}(F)$ for some $F \in \mathcal{F}$. Then $\widetilde{\pi}(\alpha_{\text{new}}) = \pi(F) = [F]_{\text{gauge}} \in \mathfrak{M}$ — an existing point. No new base point is produced. □

Corollary 8.8 (Intensional refinement is slice-level)**.** *Intensional refinement adds no new structural axis to AFN-T: the base ($\mathfrak{M}$) is untouched, and TH-absoluteness holds without modification.*

Proof. Five-axis absoluteness (Theorems 7.1–7.6) concerns points of $\mathfrak{M}$. Theorem 8.7(c) shows intensional refinement acts only on fibers, not base points. No new axis of variation is introduced. □

8.1 The Intensional Grading

The slice-locality theorem presents each fiber $\text{Int}([F])$ as a bare 2-category. The inductive construction of the canonical display class in Theorem 8.6 (Step 1) in fact endows every fiber with finer, functorial structure: an $\mathbb{N}$-indexed grading measuring *how far from an equivalence* a display datum is. This subsection records the grading and its two structural properties; both proofs are short because the generating induction of Theorem 8.6 already carries them implicitly.

Definition 8.9 (Display grade)**.** Let $F \in \mathcal{F}$ and let $\mathcal{D}_F = \bigcup_{n<\omega} \mathcal{D}_F^{(n)}$ be the canonical minimal display class with its generating filtration (Theorem 8.6, Step 1). For $d \in \mathcal{D}_F$ define the *grade*

$$\text{gr}(d) := \min\{\, n : d \in \mathcal{D}_F^{(n)} \,\} \in \mathbb{N}.$$

Thus $\text{gr}(d) = 0$ iff d is a 1-equivalence, and $\text{gr}(d) = n+1$ iff d first arises at stage $n+1$, i.e. as a 2-pullback, a composite, or a 2-cell transport of data of grade $\leq n$, but is not itself of grade $\leq n$. For an object $\mathbf{I}(F) = (\mathcal{C}_F, \mathcal{D}_F, \iota_F) \in \mathcal{S}_{\text{int}}$, write $\text{Int}^{(n)}([F]) \subseteq \text{Int}([F])$ for the full sub-2-category of the fiber (Theorem 8.7(b)) on display data of grade $\leq n$.

Proposition 8.10 (The grading is structural)**.** *(a)* Exhaustive filtration*:* $\text{Int}^{(0)}([F]) \subseteq \text{Int}^{(1)}([F]) \subseteq \cdots$ *with* $\text{Int}([F]) = \text{colim}_n \text{Int}^{(n)}([F])$.

(b) Functoriality (grade non-increase)*: for every* 1*-morphism* $f : F_1 \to F_2$ *in* $\mathcal{F}$*, the pullback component* $f^*|_{\mathcal{D}}$ *of* $\mathbf{I}(f)$ *satisfies* $\text{gr}(f^*d) \leq \text{gr}(d)$ *for all* $d \in \mathcal{D}_{F_2}$*; hence* $\mathbf{I}(f)$ *restricts to each grading stage,* $\mathbf{I}(f) : \text{Int}^{(n)}([F_2]) \to \text{Int}^{(n)}([F_1])$*.*

(c) Subadditivity*:* $\text{gr}(d_2 \circ d_1) \leq \max(\text{gr}(d_1), \text{gr}(d_2)) + 1$*, and* $\text{gr}(\phi^* d) \leq \text{gr}(d) + 1$ *for* 2*-cell transports* ϕ*.*

Proof. (a) is immediate from $\mathcal{D}_F = \bigcup_n \mathcal{D}_F^{(n)}$. (b) is the content of Theorem 8.6, Step 2: the preservation of display data under f^* is proved there *by induction on the class generation* — the base class (1-equivalences, grade 0) is preserved by pullback, and each inductive closure step is preserved stage-by-stage by the pullback-of-pullback property; reading that induction with its stage index retained yields $f^*(\mathcal{D}_{F_2}^{(n)}) \subseteq \mathcal{D}_{F_1}^{(n)}$ for every n, which is the claim. (c) is the definition of the generating step: composites and 2-cell transports of data available at stage $\max(\mathrm{gr}(d_1), \mathrm{gr}(d_2))$ (resp. $\mathrm{gr}(d)$) are enrolled at the next stage. □

Corollary 8.11 (Extensionality is grade-blindness)**.** *The Morita 2-localization of (I-4) inverts exactly the grade-0 layer: the localizing class $\mathcal{W}_{\mathcal{U}}$ consists of morphisms whose underlying 2-functor is an equivalence, and on display data these are the grade-0 elements. Consequently the extensional quotient $\mathcal{S}_{\mathrm{int}}[\mathcal{W}_{\mathcal{U}}^{-1}] \simeq \mathfrak{M}$ identifies all grades: the composite $\mathrm{Int}^{(n)}([F]) \hookrightarrow \mathrm{Int}([F]) \to \{[F]\}$ is constant in n. The intensional content of a foundation is precisely the tower $\{\mathrm{Int}^{(n)}([F])\}_n$ that the extensional projection forgets.*

Remark 8.12 (Reading, and an open question)**.** The grading gives the slice fibers of Theorem 8.7 the structure that the extensional-collapse example (Theorem 8.6, Step 7) implicitly exploits: the typing invariant τ is computed on display data of *bounded grade*, and the MLTT/ETT distinction is visible already at finite stages of the tower. Two natural questions remain open: (i) *strictness* — for which F is the filtration $\mathrm{Int}^{(n)}([F])$ strictly increasing in n (no grade collapse at a finite stage)? A collapse at stage n_0 would express a strong normal-form property of F's display geometry; (ii) *grade as a Morita-refinement invariant* — whether the minimal collapse stage $n_0(F) \in \mathbb{N} \cup \{\infty\}$, which is invariant under $\mathcal{S}_{\mathrm{int}}$-equivalence by Proposition 8.10(b), separates Morita-equivalent foundations beyond the binary invariant τ of Step 7, providing a quantitative refinement scale between extensional identity and intensional distinctness.

9 $\mathcal{L}_{\mathrm{Cls}}$ Meta-Classification

9.1 The Class $\mathfrak{Meta}_{\mathrm{Cls}}$

Definition 9.1 ($\mathcal{L}_{\mathrm{Cls}}$ meta-structure)**.** An object $\mathbf{F}$ is a $\mathcal{L}_{\mathrm{Cls}}$ **meta-structure** if it satisfies:

- **(M1) Locally small 2-category**: $\mathbf{F}$ is a locally small 2-category.
- **(M2) Classification functor**: there is a 2-functor $\mathrm{Cl}_{\mathbf{F}} : \mathbf{F} \to \mathfrak{M}$ landing in the classifying 2-stack of Rich-foundations; we write $\mathfrak{M}(\mathbf{F}) := \mathrm{image}(\mathrm{Cl}_{\mathbf{F}}) \subseteq \mathfrak{M}$ for the sub-stack carved out by $\mathbf{F}$'s classification.
- **(M3) Accessible reflection**: there is an accessible endofunctor $R_{\mathbf{F}} : \mathbf{F} \to \mathbf{F}$.
- **(M4) Non-generative**: $\mathbf{F}$ does not derive the axioms of foundations $F \in \mathrm{image}(\mathrm{Cl}_{\mathbf{F}})$ from its own axioms; classification is parametric, not axiomatic.
- **(M5) Metatheory-parametrized**: $\mathbf{F}$ is defined within some $S \in$ R-S.

I denote the class of such $\mathbf{F}$ as $\mathfrak{Meta}_{\mathrm{Cls}}$.

Definition 9.2 (Maximality)**.** A meta-structure $\mathbf{F} \in \mathfrak{Meta}_{\mathrm{Cls}}$ is **maximal** if additionally:

- **(Max-1) Full classification**: $\mathrm{image}(\mathrm{Cl}_{\mathbf{F}}) = \mathfrak{M}$.
- **(Max-2) Gauge-fullness**: $\mathrm{Aut}_2(\mathbf{F})$ acts transitively on gauge classes in $\mathrm{image}(\mathrm{Cl}_{\mathbf{F}})$.

- **(Max-3) Depth stratification**: $\mathbf{F}$ admits a *depth-indexed filtration* $\mathbf{F}^{(0)} \hookrightarrow \mathbf{F}^{(1)} \hookrightarrow \cdots \hookrightarrow \mathbf{F}^{(n)} \hookrightarrow \cdots$ of accessible 2-subcategories with $\mathbf{F} \simeq_2 \operatorname{colim}_{n<\omega} \mathbf{F}^{(n)}$, equipped with a *depth function* depth : $\mathrm{Ob}(\mathbf{F}) \to \omega$ assigning to each object the least n such that the object lies in $\mathbf{F}^{(n)}$, satisfying the following *predicativity condition*: for every predicate P appearing in a definition by comprehension of an object $A \in \mathbf{F}^{(n)}$ (i.e., the definition has the form "$A :=$ the object classifying $\{x : P(x)\}$"), every object Y that appears as the target of a sub-expression in P — in particular every target of $R_{\mathbf{F}}$, $\mathrm{Cl}_{\mathbf{F}}$, or subobject-classifier evaluations — satisfies $\mathrm{depth}(Y) < n$.

- **(Max-4) Intensional completeness with conservativity**: there exists a 2-functor $\mathbf{I}_{\mathbf{F}} : \mathbf{F}^{\mathrm{op}} \to \mathcal{S}_{\mathrm{int}}$ satisfying properties (I-1)–(I-4) of Theorem 8.6, and moreover the pair $(\mathrm{Cl}_{\mathbf{F}}, \mathbf{I}_{\mathbf{F}})$ is *jointly conservative*: two parallel 1-morphisms of $\mathbf{F}$ that are identified both by $\mathrm{Cl}_{\mathbf{F}}$ and by $\mathbf{I}_{\mathbf{F}}$ are equal (up to invertible 2-cell). The canonical classifier $\mathbf{F}^*_S$ of Theorem 9.3 satisfies this clause by construction (its morphisms are pairs of a base morphism and an intensional-fiber morphism), so the clause does not empty the class.

I denote the class of maximal meta-structures as $\mathfrak{Meta}^{\top}_{\mathrm{Cls}}$.

9.2 Meta-Categoricity

Theorem 9.3 (Meta-Categoricity)**.** *Let $\mathbf{F}_1, \mathbf{F}_2 \in \mathfrak{Meta}^{\top}_{\mathrm{Cls}}$ be parametrized by the same Rich-metatheory $S \in$ R-S. Then $\mathbf{F}_1 \simeq_{(\infty,\infty)} \mathbf{F}_2$ via a canonical equivalence.*

Proof. I show that each $\mathbf{F}_i$ is canonically (∞,∞)-equivalent to a common object $\mathbf{F}^*_S$ constructed from S, whence $\mathbf{F}_1 \simeq_{(\infty,\infty)} \mathbf{F}^*_S \simeq_{(\infty,\infty)} \mathbf{F}_2$.

Step 1: the canonical maximal classifier. Let $\rho : \mathcal{F} \to (\infty, n_S)$-Cat be the classification functor of Convention 1.2. Consider the Grothendieck straightening applied to the Cartesian fibration $\int \rho \to S$, with gauge-quotient:

$$\mathbf{F}^*_S := \mathbf{St}(\rho)/\mathrm{gauge},$$

where $\mathbf{St}(\rho)$ is the straightening of ρ.

Step 2: construction of $\Phi_{\mathbf{F}}$. Any $\mathbf{F} \in \mathfrak{Meta}^{\top}_{\mathrm{Cls}}$ carries four structures: a classification functor $\mathrm{Cl}_{\mathbf{F}} : \mathbf{F} \to \mathfrak{M}$ essentially surjective onto $\mathfrak{M}$ by (Max-1); a transitive Aut_2-action on gauge classes by (Max-2); a canonical depth filtration $\mathbf{F}^{(\bullet)}$ by (Max-3); a canonical intensional completion $\mathbf{I}_{\mathbf{F}} : \mathbf{F}^{\mathrm{op}} \to \mathcal{S}_{\mathrm{int}}$ with (I-1)–(I-4) by (Max-4).

Each structure admits a universal presentation on $\mathbf{F}^*_S$: classification is identity on $\mathfrak{M}$; gauge is canonical; the depth filtration is $\mathbf{F}^{*(n)}_S := \{[\alpha] : \mathrm{depth}^*_S([\alpha]) < n\}$ with $\mathrm{depth}^*_S([\alpha])$ the quantifier-rank of the defining formula of $F_{[\alpha]}$ in $\mathrm{Syn}(S)$ (finite by (R2), (F-3); (Max-3) predicativity follows by Kleene arithmetic-hierarchy induction); $\mathbf{I}_{\mathbf{F}^*_S}$ is the Yoneda embedding factoring through $\tilde{\pi}$ (Theorem 8.7). By the universal property of Grothendieck straightening (Lurie HTT [36] Theorem 3.2.0.1) in $\mathbf{StrCat}_{S,2}$, there exists a unique (up to canonical $(\infty,2)$-isomorphism) 2-functor

$$\Phi_{\mathbf{F}} : \mathbf{F} \to \mathbf{F}^*_S$$

preserving all four structures.

Step 3: $\Phi_{\mathbf{F}}$ is essentially surjective. For every gauge-class $[\alpha] \in \mathfrak{M}$, (Max-1) gives $A \in \mathbf{F}$ with $\mathrm{Cl}_{\mathbf{F}}(A) = [\alpha]$, and $\Phi_{\mathbf{F}}(A)$ projects to the same class in $\mathbf{F}^*_S$. Transitivity of the $\mathrm{Aut}_2(\mathbf{F})$-action on gauge classes (Max-2) ensures this covers all of $\mathbf{F}^*_S$.

Step 4: $\Phi_{\mathbf{F}}$ is fully faithful. I prove that for every $A, B \in \mathbf{F}$ the induced map

$$\Phi_{\mathbf{F}} : \mathrm{Hom}_{\mathbf{F}}(A, B) \longrightarrow \mathrm{Hom}_{\mathbf{F}^*_S}(\Phi_{\mathbf{F}}(A), \Phi_{\mathbf{F}}(B))$$

is an equivalence of Hom-spaces.

Faithfulness: suppose $f, g : A \to B$ in $\mathbf{F}$ have $\Phi_{\mathbf{F}}(f) = \Phi_{\mathbf{F}}(g)$. Applying $\mathrm{Cl}_{\mathbf{F}}$ to both sides yields $\mathrm{Cl}_{\mathbf{F}}(f) = \mathrm{Cl}_{\mathbf{F}}(g)$ in $\mathfrak{M}$, so f and g differ at most by gauge. Applying $\mathbf{I}_{\mathbf{F}}$ (which satisfies (I-1) homotopy invariance and (I-2) gauge covariance of Theorem 8.6), $\mathbf{I}_{\mathbf{F}}(f) = \mathbf{I}_{\mathbf{F}}(g)$ in $\mathcal{S}_{\mathrm{int}}$. By the joint-conservativity clause of (Max-4), the pair $(\mathrm{Cl}_{\mathbf{F}}, \mathbf{I}_{\mathbf{F}})$ separates parallel morphisms of $\mathbf{F}$. Hence $f = g$. (Joint conservativity is an axiom of maximality, not a consequence of (I-3)–(I-4) alone: (I-3) is an object-level distinction and does not by itself separate morphisms.)

Fullness: given $h : \Phi_{\mathbf{F}}(A) \to \Phi_{\mathbf{F}}(B)$ in $\mathbf{F}^*_S$, I lift h to $\mathbf{F}$. The morphism h factors as $h_{\mathrm{ext}} \in \mathrm{Hom}_{\mathfrak{M}}(\mathrm{Cl}_{\mathbf{F}}(A), \mathrm{Cl}_{\mathbf{F}}(B))$ (extensional part, via the canonical projection $\mathbf{F}^*_S \to \mathfrak{M}$) together with $h_{\mathrm{int}} \in \mathrm{Hom}_{\mathcal{S}_{\mathrm{int}}}(\mathbf{I}_{\mathbf{F}^*_S}(B), \mathbf{I}_{\mathbf{F}^*_S}(A))$ (intensional fiber, via $\mathbf{I}_{\mathbf{F}^*_S}$). By (Max-1), h_{ext} lifts to a gauge-class of morphisms $A \to B$ in $\mathcal{F}$ at the classified level; (Max-2) transitivity picks a representative in $\mathbf{F}$. By (Max-4), h_{int} lifts uniquely to the intensional completion of $\mathbf{F}$, which by (I-4) Morita-as-2-localization coincides with the intensional data of $\mathbf{I}_{\mathbf{F}}$. Combining, I obtain a morphism $\tilde{h} : A \to B$ in $\mathbf{F}$ with $\Phi_{\mathbf{F}}(\tilde{h}) = h$.

2-*morphisms*: the same argument applied one dimension up — using (M3) accessibility of the reflection and (Max-4) intensional 2-functoriality — establishes the Hom-equivalence at the level of 2-cells. The full coherent $(\infty, 2)$-equivalence follows by induction on truncation levels.

Hence $\Phi_{\mathbf{F}}$ is a 2-equivalence in $\mathbf{StrCat}_{S,2}$.

Step 5: lifting to (∞, ∞). The 2-equivalence $\Phi_{\mathbf{F}}$ established in Steps 3–4 extends levelwise: at each truncation $\tau_{\le n}$ for $n \in \mathbb{N}$, I obtain an (∞, n)-equivalence by (M3)-accessibility of all structures involved, combined with the unicity theorem of Barwick–Schommer-Pries [9]. The family $\{\tau_{\le n}(\Phi_{\mathbf{F}})\}_{n \in \mathbb{N}}$ is compatible with truncation transitions by Whitehead's criterion (Lemma A.6). Taking the colimit through the (∞, ∞)-stabilization (Theorem A.7) yields an (∞, ∞)-equivalence $\Phi_{\mathbf{F}} : \mathbf{F} \xrightarrow{\simeq_{(\infty,\infty)}} \mathbf{F}^*_S$.

Conclusion. Applying Step 5 to both $\mathbf{F}_1$ and $\mathbf{F}_2$: the composition

$$\mathbf{F}_1 \xrightarrow{\Phi_{\mathbf{F}_1}} \mathbf{F}^*_S \xrightarrow{\Phi^{-1}_{\mathbf{F}_2}} \mathbf{F}_2$$

is a canonical (∞, ∞)-equivalence. □

9.3 Structural Multiplicity

Theorem 9.4 (Structural multiplicity)**.** *In $\mathfrak{Meta}_{\mathrm{Cls}} \setminus \mathfrak{Meta}^{\top}_{\mathrm{Cls}}$, there exist at least three pairwise non-2-equivalent meta-structures:*

- $\mathbf{F}_{\mathrm{univalent}}$*: the Univalent Foundations programme (Voevodsky's* Foundations *Coq library [49]; HoTT Book [48]; Kapulkin–Lumsdaine [30]; Shulman [14] for univalent universes in every Grothendieck ∞-topos).*
- $\mathbf{F}_{\mathrm{cosmoi}}$*: the ∞-cosmoi of Riehl–Verity [42].*
- $\mathbf{F}_{\mathrm{cohesive}}$*: the cohesive higher topos framework of Schreiber [44].*

Proof. **Step 1** (each is in $\mathfrak{Meta}_{\mathrm{Cls}}$). Each of $\mathbf{F}_{\mathrm{univalent}}$, $\mathbf{F}_{\mathrm{cosmoi}}$, $\mathbf{F}_{\mathrm{cohesive}}$ satisfies (M1)–(M5) of Definition 9.1. I verify:

- **(M1)**: each is a locally small 2-category (standard from their respective definitions).
- **(M2)**: each has a classification functor whose image is a specific sub-2-stack of $\mathfrak{M}$:

$$\begin{aligned}\mathrm{image}(\mathrm{Cl}_{\mathbf{F}_{\mathrm{univalent}}}) &= \{[\alpha_F] : F \supseteq \mathsf{MLTT} + \mathrm{Univalence}\},\\ \mathrm{image}(\mathrm{Cl}_{\mathbf{F}_{\mathrm{cosmoi}}}) &= \{[\alpha_F] : F \text{ is an } (\infty,1)\text{-categorical theory}\},\\ \mathrm{image}(\mathrm{Cl}_{\mathbf{F}_{\mathrm{cohesive}}}) &= \{[\alpha_F] : F \text{ supports cohesion modalities}\}.\end{aligned}$$

- **(M3)**: each supports an accessible reflection — truncation $\tau_{\leq n}$ for $\mathbf{F}_{\mathrm{univalent}}$, homotopy reflection for $\mathbf{F}_{\mathrm{cosmoi}}$, cohesive modalities for $\mathbf{F}_{\mathrm{cohesive}}$.
- **(M4)**: none derives axioms of its classified foundations; all are parametric.
- **(M5)**: each is parametrized by an appropriate $S \in$ R-S.

Step 2 (pairwise non-equivalence). I exhibit concrete distinctions in classification images:

- $\mathbf{F}_{\mathrm{univalent}}$ vs. $\mathbf{F}_{\mathrm{cosmoi}}$: $[\alpha_{\mathrm{NCG}}] \in \mathrm{image}(\mathrm{Cl}_{\mathbf{F}_{\mathrm{cosmoi}}})$ (NCG is an $(\infty,1)$-theory via spectral triples) but $[\alpha_{\mathrm{NCG}}] \notin \mathrm{image}(\mathrm{Cl}_{\mathbf{F}_{\mathrm{univalent}}})$ (NCG is not HoTT-extension).
- $\mathbf{F}_{\mathrm{univalent}}$ vs. $\mathbf{F}_{\mathrm{cohesive}}$: standard HoTT without cohesion lies in $\mathrm{image}(\mathrm{Cl}_{\mathbf{F}_{\mathrm{univalent}}})$ but not $\mathrm{image}(\mathrm{Cl}_{\mathbf{F}_{\mathrm{cohesive}}})$.
- $\mathbf{F}_{\mathrm{cosmoi}}$ vs. $\mathbf{F}_{\mathrm{cohesive}}$: non-cohesive $(\infty,1)$-theories lie in $\mathrm{image}(\mathrm{Cl}_{\mathbf{F}_{\mathrm{cosmoi}}})$ but not $\mathrm{image}(\mathrm{Cl}_{\mathbf{F}_{\mathrm{cohesive}}})$.

Non-equivalence is asserted in $\mathfrak{Meta}_{\mathrm{Cls}}$ *over* $\mathfrak{M}$: morphisms of meta-structures are required to commute with the classification functors (up to invertible 2-cell). If any two of the three were 2-equivalent over $\mathfrak{M}$, their classification images would coincide as sub-stacks of $\mathfrak{M}$, contradicting these observations. (As bare 2-categories, abstract equivalences ignoring Cl are not excluded and are irrelevant to the plurality claim.)

Step 3 (none is maximal). None of the three satisfies (Max-1) (full classification), since their classification images are strict subsets of $\mathfrak{M}$. Therefore, $\mathbf{F}_{\mathrm{univalent}}, \mathbf{F}_{\mathrm{cosmoi}}, \mathbf{F}_{\mathrm{cohesive}} \notin \mathfrak{Meta}^{\top}_{\mathrm{Cls}}$, and Theorem 9.3 does not apply. □

Corollary 9.5 (Plurality at $\mathcal{L}_{\mathrm{Cls}}$). *$\mathcal{L}_{\mathrm{Cls}}$ is structurally plural: $\mathfrak{Meta}_{\mathrm{Cls}}$ contains multiple non-equivalent members.*

9.4 Meta-Classification Stabilization

Theorem 9.6 (Meta-classification stabilization). *Define $\mathfrak{M}^{(\mathrm{Cls})}$ as the classifying 2-stack of $\mathcal{L}_{\mathrm{Cls}}$ meta-structures:*

$$\mathfrak{M}^{(\mathrm{Cls})} := \mathfrak{Meta}_{\mathrm{Cls}}/\textit{meta-gauge}.$$

Then:

- ***(a)*** $\mathfrak{M}^{(\mathrm{Cls})} \in \mathfrak{Meta}_{\mathrm{Cls}}$ *(the classifying 2-stack is itself $\mathcal{L}_{\mathrm{Cls}}$).*
- ***(b)*** ***Theory-level idempotence:*** $\mathfrak{M}^{(\mathrm{Cls}\cdot 2)} := \mathfrak{Meta}_{\mathrm{Cls}}^{(\textit{of } \mathfrak{M}^{(\mathrm{Cls})})}/\textit{meta-gauge is equivalent to}$ $\mathfrak{M}^{(\mathrm{Cls})}$ as an (∞,∞)-theory — *i.e., they realize the same (∞,∞)-theory-moduli in the sense of Barwick–Schommer-Pries [9] unicity. The two stacks are* not *identified as set-theoretic objects: $\mathfrak{M}^{(\mathrm{Cls}\cdot 2)}$ lives one Grothendieck-universe step above $\mathfrak{M}^{(\mathrm{Cls})}$, and the meta-iteration tower $\{\mathfrak{M}^{(\mathrm{Cls}\cdot k)}\}_{k\geq 1}$ ascends the inaccessible hierarchy $\kappa_1 < \kappa_2 < \cdots$; the equivalence $\simeq_2$ records that each level instantiates the same (∞,∞)-theory, not that the levels are set-theoretically identical.*

(c) *No $\mathcal{L}_{\text{Abs}}$ escalation occurs through meta-iteration.*

Proof. **(a)**: $\mathfrak{M}^{(\text{Cls})}$ is a quotient 2-stack over $\mathfrak{Meta}_{\text{Cls}}$. Its conditions:

- **(M1)**: locally small (inherited from $\mathfrak{Meta}_{\text{Cls}}$).
- **(M2)**: classification functor is the identity on gauge classes (meta-structures classify themselves).
- **(M3)**: accessible reflection $R_{\mathfrak{M}^{(\text{Cls})}}$ is induced from $R_{\mathbf{F}}$ on components.
- **(M4)**: non-generative by inheritance.
- **(M5)**: parametrized by the same $S \in$ R-S.

Therefore $\mathfrak{M}^{(\text{Cls})} \in \mathfrak{Meta}_{\text{Cls}}$.

(b): I reduce meta-stabilization to the (∞,∞)-stabilization theorem (Theorem A.7) through a formal categorical-depth embedding, rather than by analogy.

Step (b.1): Categorical depth of $\mathfrak{Meta}_{\text{Cls}}$. Each $\mathbf{F} \in \mathfrak{Meta}_{\text{Cls}}$ is, by (M1)–(M2), a locally small 2-category together with a classification functor $\text{Cl}_{\mathbf{F}}$ into $\mathfrak{M}$. Since $\mathfrak{M}$ is an (∞,∞)-stack (Convention 1.2), and $\text{Cl}_{\mathbf{F}}$ is required to be accessible (M3), $\mathbf{F}$ lies in the (∞,∞)-category of accessible (∞,∞)-presheaves on $\mathfrak{M}$. Denote this ambient category by $\Pi_{(\infty,\infty)}$. Then $\mathfrak{Meta}_{\text{Cls}} \hookrightarrow \Pi_{(\infty,\infty)}$.

Step (b.2): Classifying stack of $\mathfrak{Meta}_{\text{Cls}}$ lives one level higher. The quotient $\mathfrak{M}^{(\text{Cls})} := \mathfrak{Meta}_{\text{Cls}}/\text{gauge}_{\text{meta}}$ is, by the Grothendieck–Lurie construction (Lurie HTT [36] §3.2), an (∞,∞)-stack over $\mathfrak{M}$; viewed as an object classifying the objects of $\Pi_{(\infty,\infty)}$, it lies in the one-higher category $\Pi_{(\infty,\infty+1)}$.

Step (b.3): Iteration adds one more level. By the same construction, $\mathfrak{M}^{(\text{Cls}\cdot 2)} := \mathfrak{Meta}_{\text{Cls}}^{(\text{of } \mathfrak{M}^{(\text{Cls})})}/\text{gauge}_{\text{meta}}$ is a classifying stack one level above $\mathfrak{M}^{(\text{Cls})}$, namely an object of $\Pi_{(\infty,\infty+2)}$.

Step (b.4): Application of (∞,∞)-stabilization. By Theorem A.7 (Barwick–Schommer-Pries [9]), the canonical inclusions

$$\Pi_{(\infty,\infty)} \hookrightarrow \Pi_{(\infty,\infty+1)} \hookrightarrow \Pi_{(\infty,\infty+2)}$$

are equivalences of (∞,∞)-categories: since accessible-functor categories preserve equivalences in the target (∞,∞)-cat (and the accessibility constraint is invariant under the $(\infty,n) \hookrightarrow (\infty,n+1)$ truncation jumps), Theorem A.7's equivalence on (∞,∞)-Cat itself transfers to the accessible-presheaf category $\Pi_{(\infty,\infty)} = \text{Fun}^{\text{acc}}(\mathfrak{M}, (\infty,\infty)\text{-Cat})$. Therefore the canonical 2-functors

$$\iota_{\text{meta}} : \mathfrak{M}^{(\text{Cls})} \hookrightarrow \mathfrak{M}^{(\text{Cls}\cdot 2)}, \quad \pi_{\text{meta}} : \mathfrak{M}^{(\text{Cls}\cdot 2)} \to \mathfrak{M}^{(\text{Cls})}$$

satisfy $\pi_{\text{meta}} \circ \iota_{\text{meta}} = \text{id}$ and $\iota_{\text{meta}} \circ \pi_{\text{meta}} \simeq_2 \text{id}$, the latter being the image of the stabilization equivalence under the Grothendieck–Lurie construction.

Step (b.5): Conclusion. $\mathfrak{M}^{(\text{Cls}\cdot 2)} \simeq_2 \mathfrak{M}^{(\text{Cls})}$, which is the claim.

(c): Suppose, for contradiction, $\mathfrak{M}^{(\text{Cls}\cdot k)}$ (for any $k \geq 1$) satisfies the $\mathcal{L}_{\text{Abs}}$ conditions $(F_S) \wedge (\Pi_{4,S,n}) \wedge (\Pi_{3\text{-max},S,n})$ (Definition 4.4). By (b), $\mathfrak{M}^{(\text{Cls}\cdot k)}$ realizes the same (∞,∞)-theory as $\mathfrak{M}^{(\text{Cls})} \in \mathfrak{Meta}_{\text{Cls}}$ (though at a higher Grothendieck-universe level κ_k). As a $\mathcal{L}_{\text{Cls}}$ meta-structure, $\mathfrak{M}^{(\text{Cls}\cdot k)}$ is classifying but not generative of Rich-foundations (M4), contradicting $\mathcal{L}_{\text{Abs}}$ maximal generativity $(\Pi_{3\text{-max},S,n})$. The contradiction shows no $\mathcal{L}_{\text{Abs}}$ escalation occurs. □

Corollary 9.7 (Iteration closure)**.** *For every $k \geq 1$, $\mathfrak{M}^{(\text{Cls}\cdot k)}$ realizes the same (∞,∞)-theory as $\mathfrak{M}^{(\text{Cls})}$ (notation $\simeq_2$ reserved for this theory-level equivalence of Theorem 9.6(b)). Meta-iteration stabilizes at $\mathcal{L}_{\text{Cls}}$* as a theory*; its set-theoretic realization ascends the Grothendieck-universe hierarchy κ_k with k.*

Corollary 9.8 (Closure of $\mathfrak{Meta}_{\mathrm{Cls}}$ under meta-classification)**.** *The class $\mathfrak{Meta}_{\mathrm{Cls}}$ is closed under classification iteration: iterated meta-classification of $\mathcal{L}_{\mathrm{Cls}}$ meta-structures remains at $\mathcal{L}_{\mathrm{Cls}}$.*

10 AC/OC Duality and the Dual Boundary Lemma

The Lambek–Scott adjunction $(\mathrm{Syn}, \mathrm{Mod})$ of (R5b), assembled across R-S and over $n \in \mathbb{N} \cup \{\infty\}$, induces a (∞,∞)-equivalence between the theory-centric moduli $\mathfrak{M}$ and the enactment-centric moduli $\mathfrak{M}_{\mathcal{E}}$ (Theorem 10.5). The dual boundary lemma $\mathcal{L}^{\mathcal{E}}_{\mathrm{Abs}} = \emptyset$ follows (Theorem 10.8).

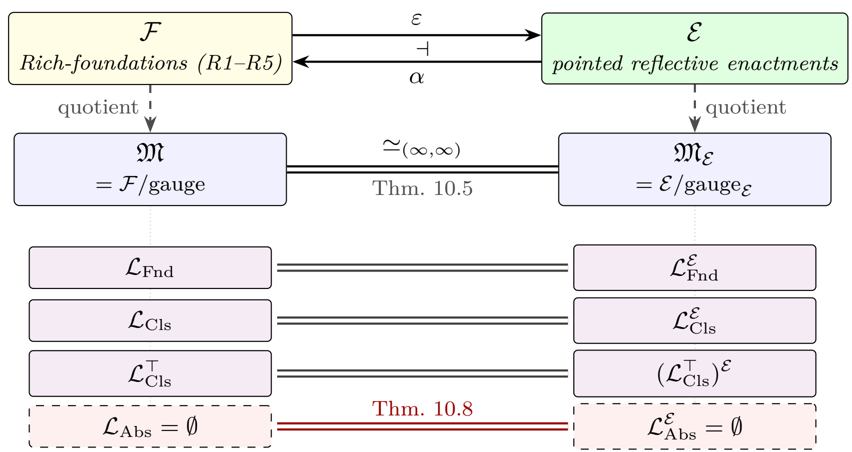

Figure 3: AC/OC duality and the induced bijection of strata. At the 2-category level (top row), the adjoint pair $\varepsilon \dashv \alpha$ (Definition 10.1) relates the theory-centric $\mathcal{F}$ to the enactment-centric $\mathcal{E}$: ε is the syntactic self-enactment section $F \mapsto (F, \mathrm{Syn}(F), \mathrm{id}, \mathrm{id})$; α is the forgetful functor $(F, \mathcal{C}, \iota, r) \mapsto F$. Passing to gauge-classes (middle row) yields an (∞,∞)-equivalence of classifying 2-stacks $\mathfrak{M} \simeq \mathfrak{M}_{\mathcal{E}}$ (Theorem 10.5), via full faithfulness of ε and essential surjectivity through the chosen reflector r. Under this equivalence the four strata correspond bijectively (bottom rows): $\mathcal{L}_{\mathrm{Fnd}} \leftrightarrow \mathcal{L}^{\mathcal{E}}_{\mathrm{Fnd}}$, $\mathcal{L}_{\mathrm{Cls}} \leftrightarrow \mathcal{L}^{\mathcal{E}}_{\mathrm{Cls}}$, $\mathcal{L}^{\top}_{\mathrm{Cls}} \leftrightarrow (\mathcal{L}^{\top}_{\mathrm{Cls}})^{\mathcal{E}}$, and $\mathcal{L}_{\mathrm{Abs}} = \emptyset \leftrightarrow \mathcal{L}^{\mathcal{E}}_{\mathrm{Abs}} = \emptyset$ (Theorem 10.8). The symmetry formalises the reconciliation of the object-centric and action-centric viewpoints on mathematical foundations: neither is primary; both describe the same classifying stack.

10.1 The 2-Category of Rich-Enactments

Definition 10.1 (2-category of enactments $\mathcal{E}$)**.** The 2-*category of Rich-enactments* $\mathcal{E}$ has:

- **Objects**: quadruples $(F, \mathcal{C}, \iota, r)$, called *pointed reflective enactments over* F, where $F \in \mathcal{F}$, $\mathcal{C} \in \mathbf{StrCat}_{S,n_F}$ (Definition 1.3) is a κ_1-accessible S-model-category with an F-model structure, $\iota : \mathrm{Syn}(F) \to \mathcal{C}$ is an F-preserving fully faithful (∞, n_F)-functor onto a replete sub-(∞, n_F)-category, and $r : \mathcal{C} \to \mathrm{Syn}(F)$ is a *chosen* left adjoint (reflector) to ι with unit $\eta_r : \mathrm{id}_{\mathcal{C}} \Rightarrow \iota \circ r$ and counit $\varepsilon_r : r \circ \iota \Rightarrow \mathrm{id}_{\mathrm{Syn}(F)}$ satisfying the triangle identities in $\mathbf{StrCat}_{S,n_F}$. The reflector r is part of the data; it is uniquely determined by ι up to unique invertible 2-cell (Riehl–Verity [42] Prop. 2.1.11; Adámek–Rosický [2] Thm. 1.26).

- **1-morphisms** $(\phi, \Phi) : (F_1, \mathcal{C}_1, \iota_1, r_1) \to (F_2, \mathcal{C}_2, \iota_2, r_2)$: a pair consisting of a 1-morphism $\phi : F_1 \to F_2$ in $\mathcal{F}$ (faithful interpretation) and an S-preserving (∞, n)-functor $\Phi : \mathcal{C}_1 \to \mathcal{C}_2$ equipped with an invertible 2-cell $\Phi \circ \iota_1 \Rightarrow \iota_2 \circ \mathrm{Syn}(\phi)$;

- **2-morphisms**: coherent natural transformations $(\phi, \Phi) \Rrightarrow (\phi', \Phi')$ compatible with the structure cells, i.e., pairs (τ, T) with $\tau : \phi \Rightarrow \phi'$ in $\mathcal{F}$ and $T : \Phi \Rrightarrow \Phi'$ satisfying the evident pentagon with $\iota_1, \iota_2, \mathrm{Syn}(\tau)$.

There is a canonical forgetful 2-functor and a canonical section:

$$\alpha : \mathcal{E} \to \mathcal{F}, \quad (F, \mathcal{C}, \iota, r) \mapsto F; \qquad \varepsilon : \mathcal{F} \to \mathcal{E}, \quad F \mapsto \big(F, \mathrm{Syn}(F), \mathrm{id}_{\mathrm{Syn}(F)}, \mathrm{id}_{\mathrm{Syn}(F)}\big).$$

By construction $\alpha \circ \varepsilon = \mathrm{id}_{\mathcal{F}}$ strictly. For readability I occasionally write $(F, \mathcal{C}, \iota)$ for $(F, \mathcal{C}, \iota, r)$ when the reflector is understood.

Definition 10.2 (Enactment gauge)**.** Let $\mathcal{R}_{\mathcal{E}} \subseteq \mathrm{Mor}_1(\mathcal{E})$ be the class of *reflection morphisms*: 1-morphisms of the form (τ, Φ) in which τ is a gauge transformation in $\mathcal{F}$ (Convention 1.2) and the composite $r_2 \circ \Phi \circ \iota_1 : \mathrm{Syn}(F_1) \to \mathrm{Syn}(F_2)$ is an (∞, n)-equivalence. The class $\mathcal{R}_{\mathcal{E}}$ contains all identities and all 1-equivalences, is closed under composition (reflectors compose along the invertible counits), is stable under the 2-pullbacks existing in $\mathcal{E}$ (componentwise, since gauge transformations and equivalences are pullback-stable), and is saturated; hence it satisfies the calculus-of-fractions conditions (BF1)–(BF5) of Lemma A.3. Define

$$\mathrm{gauge}_{\mathcal{E}} := \text{the 2-localization class generated by } \mathcal{R}_{\mathcal{E}}, \qquad \mathfrak{M}_{\mathcal{E}} := \mathcal{E}[\mathcal{R}_{\mathcal{E}}^{-1}]$$

via Pronk's bicategory of fractions (Lemma A.3). Two enactments are *gauge$_{\mathcal{E}}$-equivalent* iff they are connected by a zig-zag of reflection morphisms. In particular the reflection $(\mathrm{id}_F, r) : (F, \mathcal{C}, \iota, r) \to \varepsilon(F)$ is inverted in $\mathfrak{M}_{\mathcal{E}}$ — this, and only this, is what "passing to gauge" adds: the unit η_r of a proper reflection is *not* invertible in $\mathcal{E}$ itself.

10.2 The Enactment Syntax–Semantics Lemma

Definition 10.3 (The class $\mathcal{S}_S^{\mathcal{E}}$)**.** Given $S \in$ R-S, the class $\mathcal{S}_S^{\mathcal{E}} \subseteq \mathrm{Ob}\mathcal{E}$ of *S-definable pointed enactments* is the closure of the base class

$$\mathcal{S}_S^{\mathcal{E},\,\mathrm{base}} = \{(F', \mathcal{M}, \iota_{\mathcal{M}}, r_{\mathcal{M}}) : \mathcal{M} \models S,\ F' \in \mathcal{F},\ \iota_{\mathcal{M}} := \mathrm{ev}_{\mathcal{M}},\ r_{\mathcal{M}} := (\mathrm{ev}_{\mathcal{M}})^L\},$$

where $\mathrm{ev}_{\mathcal{M}} : \mathrm{Syn}(F') \to \mathcal{M}$ is the syntax-to-semantics evaluation of (R5b) and $(\mathrm{ev}_{\mathcal{M}})^L$ is its left adjoint (Lambek–Scott), under componentwise application of the operations of Definition 3.3: the F-component by Definition 3.3; the $\mathcal{C}$-component via the forgetful $\mathbf{StrCat}_{S,n} \to (\infty, n)$-Cat; the (ι, r)-components by functoriality of adjunctions under S-preserving (∞, n)-functors. The sub-class $\mathcal{S}_S^{\mathcal{E},\,\mathrm{global}} \subseteq \mathcal{S}_S^{\mathcal{E}}$ is the corresponding closure of $\mathcal{S}_S^{\mathcal{E},\,\mathrm{base}}$ using only those operations of Definition 3.3 defining $\mathcal{S}_S^{\mathrm{global}}$ (Kan extensions, Grothendieck constructions, derived functors).

Lemma 10.4 (Enactment syntax–semantics lemma)**.** *Let $(F, \mathcal{C}, \iota, r) \in \mathcal{E}$ be a pointed reflective enactment such that the quadruple is formally S-definable, i.e., there exist formulas $\phi_F, \psi_{\mathcal{C}}, \chi_\iota, \chi_r$ in some $F' \in \mathcal{F}$ with $F' \hookrightarrow S$ specifying $F, \mathcal{C}, \iota, r$ respectively, with the triangle identities and reflectiveness of ι provable in F'. Then $(F, \mathcal{C}, \iota, r) \in \mathcal{S}_S^{\mathcal{E},\,\mathrm{global}}$.*

Proof. By hypothesis, F is formally S-definable via ϕ_F; Lemma 3.4 gives $F \in \mathcal{S}_S^{\mathrm{global}}$. The model-category $\mathcal{C}$ is specified by $\psi_{\mathcal{C}}$ within F'; as an S-structured (∞, n)-category the underlying (∞, n)-category of $\mathcal{C}$ lies in $\mathcal{S}_S^{\mathrm{global}}$ by the same argument applied to the $\mathcal{S}_S^{\mathrm{global}}$-closure of Definition 3.3. The pointing ι and the reflector r, specified by χ_ι and χ_r in F', are morphisms between $\mathcal{S}_S^{\mathrm{global}}$-objects definable in $F' \hookrightarrow S$. The adjoint pair (ι, r) is realised in $\mathcal{S}_S^{\mathcal{E},\,\mathrm{global}}$ via the following explicit construction: ι is obtained as a Kan extension (Definition 3.3, O1) along the inclusion

$\mathrm{Syn}(F') \hookrightarrow \mathcal{C}$ specified by χ_ι; the existence of the *left* reflector r (left adjoint to ι per Definition 10.1) then follows from the adjoint-functor existence theorem for accessible functors between locally κ_S-presentable categories (Adámek–Rosický [2], the adjoint-functor existence theorem applied to the cocomplete-category case), with the triangle identities transported from χ_r. Assembled componentwise, $(F, \mathcal{C}, \iota, r) \in \mathcal{S}_S^{\mathcal{E},\,\mathrm{global}}$. □

10.3 AC/OC Morita Duality

Theorem 10.5 (AC/OC Morita Duality — formal-organizational form)**.** *The adjoint pair $\varepsilon \dashv \alpha$ of Definition 10.1 induces an (∞,∞)-equivalence of classifying 2-stacks*

$$\mathfrak{M} \ \simeq_{(\infty,\infty)} \ \mathfrak{M}_{\mathcal{E}}, \qquad \mathfrak{M}_{\mathcal{E}} := \mathcal{E}/\mathrm{gauge}_{\mathcal{E}},$$

where the left-hand moduli stack is the classifying 2-stack of $\mathcal{F}$ from Convention 1.2 and $\mathfrak{M}_{\mathcal{E}}$ is the corresponding classifying 2-stack of $\mathcal{E}$. Concretely:

- *(a) $\varepsilon : \mathcal{F} \to \mathcal{E}$ is fully faithful;*
- *(b) the unit $\mathrm{id}_{\mathcal{F}} \Rightarrow \alpha \circ \varepsilon$ is an equality; the counit $\varepsilon \circ \alpha \Rightarrow \mathrm{id}_{\mathcal{E}}$ is a gauge-equivalence, i.e., an equivalence after passing to $\mathfrak{M}_{\mathcal{E}}$;*
- *(c) the strata $\mathcal{L}_{\mathrm{Fnd}}, \mathcal{L}_{\mathrm{Cls}}, \mathcal{L}_{\mathrm{Cls}}^{\top}, \mathcal{L}_{\mathrm{Abs}}$ of $\mathfrak{M}$ correspond bijectively to strata $\mathcal{L}_{\mathrm{Fnd}}^{\mathcal{E}}, \mathcal{L}_{\mathrm{Cls}}^{\mathcal{E}}, (\mathcal{L}_{\mathrm{Cls}}^{\top})^{\mathcal{E}}, \mathcal{L}_{\mathrm{Abs}}^{\mathcal{E}}$ of $\mathfrak{M}_{\mathcal{E}}$ via the induced equivalence, preserving the meta-operations* Cls *and* Gen*.*

Proof. (a) Full faithfulness of ε. Let $F_1, F_2 \in \mathcal{F}$. Unpacking Definition 10.1, a 1-morphism $\varepsilon(F_1) \to \varepsilon(F_2)$ is a pair (ϕ, Φ) with $\phi : F_1 \to F_2$ in $\mathcal{F}$ and $\Phi : \mathrm{Syn}(F_1) \to \mathrm{Syn}(F_2)$ an S-preserving (∞, n)-functor equipped with an invertible 2-cell $\Phi \Rightarrow \mathrm{Syn}(\phi)$. The *classifying property* of syntactic categories (functorial semantics: S-preserving (∞, n)-functors $\mathrm{Syn}(F_1) \to \mathcal{C}$ correspond naturally to F_1-model structures on $\mathcal{C}$; Lambek–Scott [29] for $n = 1$, Kapulkin–Lumsdaine [30] for the type-theoretic $(\infty, 1)$-case), applied with $\mathcal{C} = \mathrm{Syn}(F_2)$, identifies S-preserving functors $\mathrm{Syn}(F_1) \to \mathrm{Syn}(F_2)$ with F_1-models in $\mathrm{Syn}(F_2)$, i.e. with interpretations $\phi : F_1 \to F_2$: every such functor arises, up to an invertible 2-cell, as $\mathrm{Syn}(\phi)$ for a unique ϕ. Hence the forgetful map of hom-2-categories

$$\mathrm{Hom}_{\mathcal{E}}(\varepsilon F_1, \varepsilon F_2) \longrightarrow \mathrm{Hom}_{\mathcal{F}}(F_1, F_2), \qquad (\phi, \Phi) \mapsto \phi,$$

is an $(\infty, 2)$-equivalence. This establishes full faithfulness of ε at every hom-2-category.

(b) Essential surjectivity at gauge level. Let $(F, \mathcal{C}, \iota, r) \in \mathcal{E}$. The reflection morphism $(\mathrm{id}_F, r) : (F, \mathcal{C}, \iota, r) \to \varepsilon(F) = (F, \mathrm{Syn}(F), \mathrm{id}, \mathrm{id})$ — whose structure cell is the invertible counit $\varepsilon_r : r \circ \iota \Rightarrow \mathrm{id}_{\mathrm{Syn}(F)}$ (left adjoint to a fully faithful functor; Riehl–Verity [42] Prop. 2.1.11) — lies in the class $\mathcal{R}_{\mathcal{E}}$ of Definition 10.2 (here $r_2 \circ \Phi \circ \iota_1 = \mathrm{id} \circ r \circ \iota \simeq \mathrm{id}_{\mathrm{Syn}(F)}$). By construction of $\mathfrak{M}_{\mathcal{E}} = \mathcal{E}[\mathcal{R}_{\mathcal{E}}^{-1}]$, this morphism becomes invertible in $\mathfrak{M}_{\mathcal{E}}$, yielding a canonical $\mathrm{gauge}_{\mathcal{E}}$-equivalence

$$(F, \mathcal{C}, \iota, r) \ \simeq_{\mathrm{gauge}_{\mathcal{E}}} \ \varepsilon(F) \quad \text{in } \mathfrak{M}_{\mathcal{E}}.$$

We emphasise what is *not* claimed: the unit $\eta_r : \mathrm{id}_{\mathcal{C}} \Rightarrow \iota \circ r$ is not invertible in $\mathcal{E}$ unless ι is essentially surjective; the equivalence above is created by the localization, exactly as ring-Morita equivalence is created by inverting bimodule equivalences. Uniqueness of r up to unique invertible 2-cell given ι makes the equivalence natural in the object. Hence ε is essentially surjective after passing to gauge-classes, and the theorem acquires its precise content: *the enactment moduli, localized at reflections, is equivalent to the theory moduli.*

Combined with (a), the pair (ε, α) induces an equivalence $\mathfrak{M} \simeq \mathfrak{M}_{\mathcal{E}}$ at the $(\infty, 2)$-level.

Lift to (∞,∞). The argument above is parametric in $n \in \mathbb{N} \cup \{\infty\}$: it uses only (R5b), the 2-functoriality of Syn over n, and the uniqueness of reflectors at (∞, n). The first is (R5b); the second is Kapulkin–Lumsdaine [30] combined with Barwick–Schommer-Pries [9]; the third is Adámek–Rosický [2] Thm. 1.26 applied slice-wise via Lurie HTT [36] §5.4 (λ-filtered-colimit preservation by accessible adjunctions). Stabilization of the equivalence at $n = \infty$ follows from Theorem A.7.

(c) Stratum correspondence. The classification meta-operation Cls on $\mathfrak{M}$ sends $F \in \mathcal{L}_{\mathrm{Fnd}}$ to its class of classifiable sub-stacks. Under the equivalence $\mathfrak{M} \simeq \mathfrak{M}_{\mathcal{E}}$, the enactment $\varepsilon(F) = (F, \mathrm{Syn}(F), \mathrm{id}, \mathrm{id})$ inherits the classification structure via $\iota = \mathrm{id}$, i.e., $\mathrm{Cls}^{\mathcal{E}}(\varepsilon F)$ coincides with ε applied to $\mathrm{Cls}(F)$. Analogously for Gen. The strata $\mathcal{L}^{\mathcal{E}}_{\mathrm{Cls}} \supsetneq (\mathcal{L}^{\top}_{\mathrm{Cls}})^{\mathcal{E}} \supseteq \mathcal{L}^{\mathcal{E}}_{\mathrm{Abs}}$ on $\mathfrak{M}_{\mathcal{E}}$ are defined by the $\mathcal{E}$-analogues of Definition 2.1: a gauge-class $[(F, \mathcal{C}, \iota, r)] \in \mathcal{L}^{\mathcal{E}}_{\mathrm{Cls}}$ iff $[F] \in \mathcal{L}_{\mathrm{Cls}}$; similarly for the finer strata. Under this translation, the equivalence of (a)–(b) restricts to a bijection of strata. $\square$

Corollary 10.6 (Conservativity; consistency strength)**.** *The extension of the MSFS framework by $\mathcal{E}$ and the structure maps (α, ε) is a conservative extension of the $\mathcal{F}$-based framework: every statement about $\mathfrak{M}_{\mathcal{E}}$ is equivalent, via Theorem 10.5, to a statement about $\mathfrak{M}$. In particular the ambient metatheory needed to construct and reason about $\mathcal{E}$ is the same as for $\mathcal{F}$:*

$$\mathrm{Con}(\textit{MSFS with } \mathcal{E}) \;=\; \mathrm{Con}(\textit{MSFS}) \;=\; \mathrm{Con}(\mathsf{ZFC} + 2\text{-}\textit{inacc}).$$

Proof. By Theorem 10.5, every statement about $\mathfrak{M}_{\mathcal{E}}$ translates to an equivalent statement about $\mathfrak{M}$ via the equivalence. No new axioms are introduced; by construction, $\mathcal{E}$ is $\mathbf{U}_2$-small and is constructed inside $\mathsf{ZFC} + 2$-inacc from $\mathcal{F}$ and $\mathbf{StrCat}_{S,n}$ (Definition 1.3) using only Lambek–Scott adjunctions guaranteed by (R5b) and reflectors given as object data (Definition 10.1). $\square$

The equivalence of Theorem 10.5 transfers the $\mathcal{L}_{\mathrm{Abs}}$ conditions of §4 to the enactment side. Informally, an *enactment-absolute* triple $(F, \mathcal{C}, \iota)$ is one whose enactment-side description simultaneously admits formal specification, admits no non-trivial coordination with other enactments in the S-definable class, and is maximally receptive (every other enactment factors through it).

Definition 10.7 ($\mathcal{L}^{\mathcal{E}}_{\mathrm{Abs}}$ conditions)**.** Given $S \in$ R-S and $n \in \mathbb{N} \cup \{\infty\}$, a pointed reflective enactment $(F, \mathcal{C}, \iota, r) \in \mathcal{E}$ is **enactment-absolute** if it satisfies the triple:

($\tilde{F}_S$) *Enactment-definability*: the quadruple $(F, \mathcal{C}, \iota, r)$ is formally S-definable in the sense of Lemma 10.4, i.e., specified by formulas $(\phi_F, \psi_{\mathcal{C}}, \chi_\iota, \chi_r)$ in some $F' \in \mathcal{F}$ with $F' \hookrightarrow S$ (this refines (F_S) by additionally specifying the model-category presentation, the pointing, and the reflector).

($\tilde{\Pi}_{4,S,n}$) *Non-coordination*: there is no 1-morphism $(\phi, \Phi) : (F, \mathcal{C}, \iota, r) \to (F', \mathcal{C}', \iota', r')$ in $\mathcal{E}$ with target quadruple in $\mathcal{S}^{\mathcal{E}}_S$ (Definition 10.3) such that Φ is an (∞, n)-equivalence onto its image.

($\tilde{\Pi}_{3\text{-max},S,n}$) *Maximal receptivity*: for every enactment $(F', \mathcal{C}', \iota', r') \in \mathcal{E}$ there exists a 1-morphism $(\phi, \Phi) : (F', \mathcal{C}', \iota', r') \to (F, \mathcal{C}, \iota, r)$ realizing a Morita reduction at the (∞, n)-level.

The $\mathcal{L}^{\mathcal{E}}_{\mathrm{Abs}}$ *stratum* is the class of gauge$_{\mathcal{E}}$-equivalence classes in $\mathfrak{M}_{\mathcal{E}}$ with an enactment-absolute representative.

10.4 Dual Boundary Lemma

Theorem 10.8 (Dual Boundary Lemma; Dual-AFN-T (α)-part)**.** *For every Rich-metatheory $S \in$ R-S and every categorical level $n \in \mathbb{N} \cup \{\infty\}$, no pointed reflective enactment $(F, \mathcal{C}, \iota, r) \in \mathcal{E}$ satisfies the $\mathcal{L}^{\mathcal{E}}_{\mathrm{Abs}}$ triple $(\tilde{F}_S) \wedge (\tilde{\Pi}_{4,S,n}) \wedge (\tilde{\Pi}_{3\text{-}max,S,n})$. Already the sub-pair $(\tilde{F}_S) \wedge (\tilde{\Pi}_{4,S,n})$ is structurally self-contradictory.*

Proof. Suppose $(F, \mathcal{C}, \iota, r) \in \mathcal{E}$ satisfies $(\tilde{F}_S) \wedge (\tilde{\Pi}_{4,S,n})$. By Lemma 10.4, $(F, \mathcal{C}, \iota, r) \in \mathcal{S}_S^{\mathcal{E},\,\text{global}} \subseteq \mathcal{S}_S^{\mathcal{E}}$. The identity 1-morphism $(\mathrm{id}_F, \mathrm{id}_{\mathcal{C}})$ has $\Phi = \mathrm{id}_{\mathcal{C}}$ an (∞, n)-equivalence onto its image, with target in $\mathcal{S}_S^{\mathcal{E}}$. This coordination is forbidden by $(\tilde{\Pi}_{4,S,n})$. Contradiction. Hence $\mathcal{L}_{\text{Abs}}^{\mathcal{E}} = \emptyset$. □

Corollary 10.9 ($\mathcal{L}_{\text{Abs}}^{\mathcal{E}}$ emptiness)**.** $\mathcal{L}_{\text{Abs}}^{\mathcal{E}} = \emptyset$.

Theorem 10.10 (Dual five-axis absoluteness)**.** *Theorems 7.1–7.6 transfer to $\mathcal{L}_{\text{Abs}}^{\mathcal{E}}$: the emptiness of $\mathcal{L}_{\text{Abs}}^{\mathcal{E}}$ is absolute along the five structural axes*

horizontal *(over* R-S*)*, vertical *(over n)*, meta-vertical*,* lateral*,* completeness *(within Lawvere-scope).*

In particular, the three standard bypass paths (Theorems 8.1, 8.2, 8.7) foreclose their $\mathcal{E}$-analogues.

Proof. Each axis transfers via Theorem 10.5 componentwise: the five mechanisms of Theorems 7.1–7.6 act parametrically on the F-component or naturally on the $\mathcal{C}$-component of $(F, \mathcal{C}, \iota, r)$, and the pair (α, ε) preserves them. Lawvere-scope $\mathrm{LS}(\mathcal{E})$ is defined as enactments with $F \in \mathrm{LS}(\mathcal{F})$ and $\mathcal{C}$ closed symmetric monoidal (Cartesian-closed or $*$-autonomous, covering linear logic and quantum-categorical enactments); Yanofsky's [51] generalized diagonal applies in $\mathcal{C}$ and transports to $\mathrm{Syn}(F)$ via $\iota \dashv r$. The three bypass paths transfer analogously via the componentwise action of (α, ε). □

11 Placement in the No-Go Series

AFN-T occupies a position in the structural series of foundational no-go theorems. All such results instantiate a single categorical schema: *a system sufficiently expressive to formulate a candidate maximal object is unable to contain that object without contradiction*, realized through a self-referential embedding that turns the purported maximal object into an element of the very class it was supposed to transcend. Each specializes the schema through a distinct maximality aspect and diagonal step:

Theorem	Year	Maximality aspect	Diagonal step
Cantor [10]	1891	cardinal	$x \notin f(x)$ for $f : X \to \mathcal{P}(X)$
Russell [43]	1903	comprehension	$R = \{x : x \notin x\}$, $R \in R \iff R \notin R$
Gödel (I, II) [17]	1931	completeness / consistency	$G =$ "G is not provable"
Tarski [47]	1936	semantic truth	"this sentence is not true"
Lawvere FP [35]	1969	categorical fixed-point	cartesian-closed + weakly surjective ⇒ every f has a fixed point
Ernst [12]	2015	unlimited category theory	Lawvere diagonal at $n = 1$ under Feferman (R1)–(R3)
AFN-T	—	foundational (definable + non-reducible + generative)	syntax–semantics bridge: $(F_S) \Rightarrow X \in \mathcal{S}_S^{\text{global}}$ via Lemma 3.4, contradicting (Π_4) by id_X (Theorem 5.1)
Dual-AFN-T	—	enactment-absolute (definable + non-coordination + receptive)	identity enactment $(\mathrm{id}_F, \mathrm{id}_{\mathcal{C}})$ with target in $\mathcal{S}_S^{\mathcal{E}}$ (Lemma 10.4) contradicts $(\tilde{\Pi}_4)$ (Theorem 10.8); equivalently via α + AFN-T

11.1 Subsumption

Theorem 11.1 (Subsumption via specialization (framework-level))**.** *For each classical no-go theorem T below, there exists a triple (S_T, n_T, X_T) with $S_T \in$ R-S, $n_T \in \mathbb{N} \cup \{\infty\}$, and X_T the purported maximal object of T, such that (i) $X_T \models (F_{S_T})$; (ii) X_T under the hypothesis of T would satisfy (Π_{4,S_T,n_T}); (iii) Theorem 5.1 applied at (S_T, n_T) yields a structural contradiction with the existence of X_T as $\mathcal{L}_{\text{Abs}}$-candidate. The theorem provides the* framework *in which each classical no-go inhabits; the row-specific diagonals (Cantor's $|X| < |\mathcal{P}(X)|$, Russell's $R \in R \leftrightarrow R \notin R$, etc.) are independent classical proofs that refine the abstract id_{X_T}-template into the specific witness used by T, and are not derived from AFN-T α alone.*

T	S_T	X_T	n_T	(F_{S_T})*-witness*
Cantor [10]	ZFC	$\mathcal{P} : \mathbf{Set} \to \mathbf{Set}$ *universal*	1	$\phi_{\mathcal{P}}(X, Y) \equiv Y = \mathcal{P}(X)$
Russell [43]	ZFC	*set of* $\{x : x \notin x\}$	1	$\phi_R(x) \equiv x \notin x$
Gödel I [17]	PA	G *(unprovable sentence)*	1	$\phi_G \equiv \neg\mathrm{Prov}(\ulcorner G \urcorner)$
Gödel II [17]	PA	*internal* Con(PA)*-proof*	1	ϕ_{Con} *via* Prov
Tarski [47]	PA	*arithmetical truth predicate*	1	$T(\phi) \equiv \phi$
Lawvere FP [35]	*CCC ext. of* ZFC	*w. surj.* X, $f : X \to X^X$	1	ϕ_X *in CCC-syntax*
Ernst [12]	*Feferman (R1)–(R3)*	*UCT object*	1	*Feferman unlim. schema*

In each row, AFN-T at (S_T, n_T) yields $\mathcal{L}_{\text{Abs}}(S_T, n_T) = \emptyset$ at the row's maximality aspect; the classical diagonal of T is the specific witness refinement establishing T within the framework.

Proof. Each row's X_T is S_T-definable by the listed formula (i), so $X_T \in \mathcal{S}^{\text{global}}_{S_T}$ by Lemma 3.4. $\mathrm{id}_{X_T} : X_T \to X_T$ violates (Π_{4,S_T,n_T}) per Theorem 5.1, so $X_T \notin \mathcal{L}_{\text{Abs}}$. The row-specific diagonal reconstructs: Cantor by $|X| < |\mathcal{P}(X)|$ refining $\mathrm{id}_{\mathcal{P}}$; Russell by $R \in R \leftrightarrow R \notin R$ refining id_R; Gödel by G-sentence $\leftrightarrow \neg\mathrm{Prov}(\ulcorner G \urcorner)$; Tarski's liar; Lawvere's fixed-point formula $f(x_0) = x_0$ via $\lambda x. f(x)(x)$; Ernst by Feferman unlimited-axiom schema. Each refinement specializes the universal id_{X_T} template of Theorem 5.1 to the row-specific formal witness. □

12 Consequences and Open Questions

12.1 Consequences for Foundational Programmes

AFN-T imposes a uniform structural bound on every current foundational programme: each inhabits $\mathcal{L}_{\text{Fnd}}$, or $\mathcal{L}_{\text{Cls}}$ (possibly with membership in the maximal sub-class $\mathcal{L}^{\top}_{\text{Cls}} \subseteq \mathcal{L}_{\text{Cls}}$), with $\mathcal{L}_{\text{Abs}}$ empty. Specifically:

- **Univalent Foundations** (Voevodsky [49]; HoTT Book [48]) is a $\mathcal{L}_{\text{Cls}}$ partial meta-framework (Theorem 9.4) classifying HoTT-extensions; it cannot be elevated to $\mathcal{L}_{\text{Abs}}$.
- **Higher topos theory** (Lurie [36]) and **cohesive framework** (Schreiber [44]) occupy Levels 5 and 5+ respectively.
- **Condensed mathematics** (Clausen–Scholze [11]), **perfectoid geometry**, and **∞-cosmoi** (Riehl–Verity) are bounded similarly; no $\mathcal{L}_{\text{Abs}}$ escalation is possible.
- **Proof assistants** (Coq, Lean, Agda) implement $\mathcal{L}_{\text{Fnd}}$ foundations (CIC, CIC + Univalence, MLTT); the plurality of such systems is the plurality of $\mathcal{L}_{\text{Fnd}}$.
- **Categorical logic**: classification of foundational systems by (∞, n)-categories and their meta-structures has an upper bound at $\mathcal{L}_{\text{Cls}}$, aligning with Barwick–Schommer-Pries [9] unicity.

Diagnostic: which $\mathcal{L}_{\mathrm{Abs}}$ condition fails for specific candidates? For each candidate X that has been put forward in the literature as a candidate for the absolute meta-foundational role, the precise failure mode within AFN-T's formal framework is identified by checking which member of the triple $(F_S) \wedge (\Pi_{4,S,n}) \wedge (\Pi_{3\text{-max},S,n})$ is violated.

- *Univalent Foundations (UF)*: passes (F_S) (formally definable as HoTT + univalence axiom schema), passes $(\Pi_{4,S,n})$ conditionally (not Morita-reducible to ZFC-internal structures, in the forward direction), but *fails* $(\Pi_{3\text{-}max,S,n})$: UF does not classify Rich-foundations outside the HoTT-adjacent family (e.g., it does not classify ZFC-specific cardinal arithmetic or noncommutative-geometric spectral triples by Morita reduction). Hence UF $\in \mathcal{L}_{\mathrm{Cls}} \setminus \mathcal{L}_{\mathrm{Cls}}^{\top}$.
- *Higher topos theory (Lurie)*: passes (F_S), fails $(\Pi_{3\text{-max},S,n})$: classifies categorical structures of HoTT-compatible shape but does not admit a classification functor onto Rich-foundations outside that class (e.g. first-order set theories without a HoTT-style internal language).
- *Cohesive framework (Schreiber)*: passes (F_S), fails $(\Pi_{3\text{-max}})$; additionally the cohesion axioms specialize to geometric contexts, further restricting the classification image.
- *∞-cosmoi (Riehl–Verity)*: passes (F_S) (axiomatized in ZFC + 2-inacc), fails $(\Pi_{3\text{-max}})$; they classify $(\infty, 1)$-categorical structures, not the full $\mathfrak{M}$.
- *Any hypothetical candidate X satisfying* $(F_S) \wedge (\Pi_{3\text{-}max})$: by Theorem 5.1, X then fails $(\Pi_{4,S,n})$: if X were both formally definable (via (F_S)) and maximally generative (via $(\Pi_{3\text{-max}})$), then by Lemma 3.4 $X \in \mathcal{S}_S^{\mathrm{global}}$ and id_X would realize a Morita reduction to an $\mathcal{S}_S$-target, contradicting $(\Pi_{4,S,n})$.

Thus the three conditions cannot all simultaneously hold: current candidates fail $(\Pi_{3\text{-max}})$; any hypothetical stronger candidate satisfying $(F_S) \wedge (\Pi_{3\text{-max}})$ would fail $(\Pi_{4,S,n})$. The diagnostic identifies the precise obstruction in each case rather than merely asserting impossibility.

12.2 Open Questions

(Q1) Existence of a maximal meta-framework. Theorem 9.3 asserts categoricity of $\mathfrak{Meta}_{\mathrm{Cls}}^{\top}$ conditional on its non-emptiness. Is there a concrete $\mathbf{F} \in \mathfrak{Meta}_{\mathrm{Cls}}^{\top}$ constructible in ZFC + 2-inacc? A positive answer would identify the canonical $\mathcal{L}_{\mathrm{Cls}}$ meta-framework up to (∞, ∞)-equivalence. The construction of such a witness lies outside the scope of this preprint; a candidate framework is the subject of separate ongoing work. Non-emptiness of $\mathfrak{Meta}_{\mathrm{Cls}}^{\top}$ is an existential question with *four independent structural obstacles*, corresponding to the four maximality conditions: (Max-1) requires a classifier with classification essentially surjective onto the entire $\mathfrak{M}$; (Max-2) requires a transitive Aut_2-action on gauge classes; (Max-3) requires ω-depth predicative filtration satisfying the comprehension-schema condition; and (Max-4) requires a canonical intensional completion $\mathbf{I}_{\mathbf{F}} : \mathbf{F}^{\mathrm{op}} \to \mathcal{S}_{\mathrm{int}}$ satisfying (I-1)–(I-4). Of these, (Max-4) presents a distinct existential obstacle separate from (Max-1)–(Max-3): the construction of $\mathbf{I}_{\mathbf{F}}$ as a canonical completion (not merely existing, but canonically determined by the classifier structure) is non-trivial even granting the first three conditions. Resolution of Q1 accordingly reduces to the independent construction of each of the four structural data on a common underlying classifier, not to a single sufficient condition.

(Q2) Completeness of the meta-framework list. How complete is the list $\{\mathbf{F}_{\mathrm{univalent}}, \mathbf{F}_{\mathrm{cosmoi}}, \mathbf{F}_{\mathrm{cohesive}}, \mathbf{F}_{\mathrm{HA}}, \mathbf{F}_{\mathrm{synthetic}}\}$ of currently identified $\mathcal{L}_{\mathrm{Cls}}$ meta-structures ($\mathbf{F}_{\mathrm{HA}}$ denotes Lurie's Higher Algebra programme [37])? Are there further natural meta-frameworks?

(Q3) Exhaustiveness of bypass paths. Theorems 8.1, 8.2, 8.6, 8.7 close three standard bypass paths, and Appendix B treats the paraconsistent case. Candidates for additional independent paths include modal or temporal logic extensions, substructural logics without exponential !, realizability and constructive-only foundations, and foundations grounded in non-classical computability models. Each of these appears to reduce to one of the closed paths within R-S; a formal exhaustiveness theorem is not given here and remains open.

(Q4) Consistency-strength minimality. The proof uses $\mathsf{ZFC} + 2$-inacc (Convention 1.1). Can a weaker consistency strength suffice for the full theorem, or does the meta-classification stabilization (Theorem 9.6) genuinely require the second inaccessible?

(Q5) Sub-stack of weak Rich-metatheories. Do bounded-arithmetic theories ($\mathsf{I}\Delta_0$, S_2^i) satisfying (R2)–(R5) and a restricted (R1) form a separate stratum $\mathcal{L}_{\mathrm{Fnd}}^{\mathrm{weak}} \subset \mathcal{L}_{\mathrm{Fnd}}$, or are they Morita-dense in $\mathcal{L}_{\mathrm{Fnd}}$ via reverse mathematics [28]?

A Categorical Preliminaries

I collect here the categorical tools and standard theorems invoked in the proofs above.

A.1 2-Categories

A 2**-category** $\mathcal{C}$ consists of:

- A class of objects $\mathrm{Ob}(\mathcal{C})$.
- For each pair $A, B \in \mathrm{Ob}(\mathcal{C})$, a small category $\mathrm{Hom}_{\mathcal{C}}(A, B)$ of 1-morphisms and 2-morphisms.
- Composition functors and identity 1-morphisms satisfying standard 2-categorical coherence.

A 2-category is **locally small** if each $\mathrm{Hom}_{\mathcal{C}}(A, B)$ is a small category. All 2-categories in this paper are locally small unless stated otherwise.

Reference: Kelly [33] for enriched and 2-categorical foundations; Bénabou [7] for bicategories.

A.2 (∞, n)-Categories

An (∞, n)-category is a generalization of category theory where:

- Morphisms have k-morphisms for $k \leq n$.
- k-morphisms for $k > n$ are all invertible (equivalences).

Models include: quasi-categories, Segal n-categories, complete Segal spaces, Θ_n-spaces. Equivalence of these models is established by Ara–Maltsiniotis [3].

The (∞, n)-hierarchy stabilizes at $n = \infty$: the canonical inclusion (∞, ∞)-Cat $\hookrightarrow$ $(\infty, \infty+1)$-Cat is an equivalence (Barwick–Schommer-Pries [9], applied iteratively over the Bergner–Rezk [8] comparison of models).

Reference: Lurie HTT [36] for $(\infty, 1)$-categories; Riehl–Verity [42] for synthetic $(\infty, 1)$-category theory.

A.3 Accessible Functors

A functor $F : \mathcal{C} \to \mathcal{D}$ between locally small categories is **λ-accessible** (for a regular cardinal λ) if F preserves λ-filtered colimits.

A category $\mathcal{C}$ is **locally λ-presentable** if it is cocomplete and has a small dense subcategory of λ-presentable objects.

Reference: Adámek–Rosický [2] for accessibility theory.

Lemma A.1 (Lair–Makkai–Paré representation)**.** *Every accessible category is* 2*-categorically equivalent to the* 2*-category of models of a small sketch; equivalently, to the* 2*-category of models of an accessible* 2*-theory.*

Reference for the proof. Lair [31] (1-categorical sketches, original French source); Makkai–Paré [32] Chapter 3 (systematic development, including the 2-categorical sketch/theory correspondence); Adámek–Rosický [2] Chapter 2 (accessibility-theoretic treatment). The lemma is used in this paper only through these references.

A.4 Display Map Categories

Definition A.2 (Display map category, 1-categorical version)**.** A **display map category** is a pair $(\mathcal{C}, \mathcal{D})$ where $\mathcal{C}$ is a category with pullbacks and $\mathcal{D} \subseteq \mathrm{Mor}(\mathcal{C})$ is a class of morphisms stable under pullback and containing isomorphisms.

Jacobs [21] establishes that display map categories provide categorical semantics for intensional type theories, with display morphisms corresponding to dependent type families.

Gambino–Garner [16] extends this to the 2-categorical setting, yielding Definitions 8.3 and 8.4.

A.5 Bicategory of Fractions

Lemma A.3 (Pronk's bicategory of fractions)**.** *Given a* 2*-category* $\mathcal{C}$ *and a class* $\mathcal{W}$ *of* 1*-morphisms satisfying the Bousfield–Pronk calculus-of-fractions conditions (BF1)–(BF5), the* 2*-localization* $\mathcal{C}[\mathcal{W}^{-1}]$ *exists as a bicategory of fractions.*

Reference: Pronk [40].

The conditions (BF1)–(BF5) are:

(BF1) Contains identities.

(BF2) Closed under composition.

(BF3) 2-pullback stable.

(BF4) 2-right-filtered (any morphism can be extended).

(BF5) $\mathcal{W}$-saturation: contains all 2-equivalences.

A.6 Lawvere Fixed-Point Theorem

Lemma A.4 (Lawvere fixed-point, 2-categorical)**.** *In a* 2*-cartesian closed category* $\mathcal{C}$ *with a weakly point-surjective* 1*-morphism* $e : A \to Y^A$*, every* $f : Y \to Y$ *has a fixed point: there exists* $y_0 : 1 \to Y$ *with* $f \circ y_0 = y_0$ *up to coherent* 2*-equivalence.*

Reference: Lawvere [35]; Yanofsky [51] for 2-categorical formulation.

Lemma A.5 (Lawvere fixed-point, (∞,∞)-categorical)**.** *Let $\mathcal{C}$ be an (∞,∞)-cartesian closed category admitting an object A and a 1-morphism $e : A \to Y^A$ that is* weakly point-surjective up to coherent homotopy *(every global point $1 \to Y^A$ factors through e up to invertible higher cells). Then every $f : Y \to Y$ has an (∞,∞)-coherent fixed point.*

This is obtained by extending Lemma A.4 through the Θ_n-space model of Rezk [6], passed to the limit $n \to \infty$ via the unicity theorem of Barwick–Schommer-Pries [9]. The (∞,∞)-coherent fixed point is the canonical limit of the compatible family of fixed points constructed at each finite n.

A.7 Whitehead-Type Criterion

Lemma A.6 (Whitehead-type equivalence criterion)**.** *Let $\Phi : \mathcal{C}_1 \to \mathcal{C}_2$ be an (∞,∞)-functor between accessible (∞,∞)-categories. Then Φ is an (∞,∞)-equivalence if and only if its truncations $\tau_{\leq n}(\Phi) : \tau_{\leq n}(\mathcal{C}_1) \to \tau_{\leq n}(\mathcal{C}_2)$ are (∞,n)-equivalences for every $n \in \mathbb{N}$.*

Reference: Lurie HTT [36] §6.5.4 ("Whitehead's Theorem" for $(\infty,1)$-topoi) and §5.5.6 (truncated objects in $(\infty,1)$-categories); Barwick–Schommer-Pries [9] for the (∞,n)-extension.

A.8 Bergner–Lurie Stabilization

Theorem A.7 ((∞,∞)-stabilization)**.** *Write (∞,∞)-Cat $:= \mathrm{colim}_{n<\infty}(\infty,n)$-Cat along the canonical inclusions, and define $(\infty,\infty{+}1)$-Cat as the (∞,∞)-category of categories enriched in (∞,∞)-Cat (one further enrichment step). Then the canonical inclusion (∞,∞)-Cat $\hookrightarrow (\infty,\infty+1)$-Cat is an equivalence: the colimit is idempotent under enrichment.*

Proof (assembly). At each finite level, categories enriched in (∞,n)-Cat are $(\infty,n+1)$-categories, uniquely up to equivalence by the unicity theorem of Barwick–Schommer-Pries [9] with the model comparisons of Bergner–Rezk [8]. Enrichment commutes with the filtered colimit over n, so $(\infty,\infty+1)$-Cat $\simeq \mathrm{colim}_n(\infty,n+1)$-Cat $= (\infty,\infty)$-Cat; the globular model of Ara–Maltsiniotis [3] realizes the same idempotence for ω-categories. This "unbounded-depth" reading matches the convention fixed in (R5); no statement in the paper requires the alternative everywhere-non-invertible reading. □

B Paraconsistent Extension

AFN-T as stated (Theorem 5.1) applies to classical Rich-metatheories. This appendix extends the result to paraconsistent Rich-metatheories *with extractable classical kernel* — the class of paraconsistent logics in which every theorem admits a classical-inference witness. The extension does *not* cover strongly-dialetheic theories (see below).

B.1 Paraconsistent R-S

Definition B.1 (Paraconsistent R-S with extractable classical kernel)**.** A theory S' is a **paraconsistent Rich-metatheory with extractable classical kernel** if it satisfies the classical conditions (R1')–(R5') analogous to (R1)–(R5), together with:

- **(Explosion-min)**: $\perp \nvdash \phi$ for every formula ϕ (contradictions do not entail everything).
- **(Strong-neg)**: there exists a negation $\sim$ such that $(p \wedge \sim p) \vdash_{\sim} \perp$ (a specific contradiction principle, though not full explosion).

- **(Classical-kernel extractability)**: the largest fragment $S'^{*} \subseteq S'$ closed under classical inference rules is itself consistent, non-empty, and a Rich-metatheory in R-S. Equivalently: every theorem of S' of foundational significance (in particular, the defining formula of any $\mathcal{L}_{\mathrm{Abs}}$ candidate) admits a witness derivable purely by classical inference from S''s classical-only axioms.

B.2 Paraconsistent AFN-T

Theorem B.2 (Paraconsistent AFN-T). *If S' is a paraconsistent Rich-metatheory with extractable classical kernel in the sense of Definition B.1, then AFN-T holds for S' with the same formal content as Theorem 5.1.*

Proof (sketch). Let $S'^{*} \subseteq S'$ denote the *classical kernel* of S': the largest fragment of S' closed under classical inference rules, in which the (Explosion-min) restriction is never invoked. Equivalently, $S'^{*} = \{\phi \in S' : S' \vdash \phi \text{ via classical rules only}\}$. Standard results on the Routley–Meyer star semantics (Priest [39]) and on the classical recapture for relevant logics with (Strong-neg) show that S'^{*} is well-defined, consistent, and itself a Rich-metatheory in R-S: it satisfies (R1)–(R5) because S' does, and the restriction to classical inference preserves r.e. axiomatization, model existence, Gödel coding, and categorical semantics.

Suppose AFN-T fails in S': there exists X satisfying the $\mathcal{L}_{\mathrm{Abs}}$ pair $(F_{S'}) \wedge (\Pi_{3\text{-max},S',n})$. Taking the classical projection of X — the reduct X^{*} of X to S'^{*} obtained by discarding paraconsistent inferences — yields a putative $\mathcal{L}_{\mathrm{Abs}}$ witness in S'^{*}.

Check that both conditions are preserved under classical projection:

- $(F_{S'^{*}})(X^{*})$: the defining formula ϕ_X uses only classical connectives plus $\sim$; stripping $\sim$-occurrences preserves classical definability.

- $(\Pi_{3\text{-max},S'^{*},n})(X^{*})$: every $F \in \mathcal{F}_{S'^{*}}$ is a fortiori in $\mathcal{F}_{S'}$; hence the Morita reduction $F \to X$ exists, and its classical projection yields $F \to X^{*}$.

Thus X^{*} is a $\mathcal{L}_{\mathrm{Abs}}$ witness in $S'^{*} \in$ R-S, contradicting Theorem 5.1.

Therefore AFN-T holds in S'. Full details (including the existence of the classical kernel for any paraconsistent Rich-metatheory with (Strong-neg), and the functoriality of classical projection on $\mathcal{S}_{S'}$) appear in a companion note. □

C Verum Verification Ledger

Every theorem, lemma, definition, and corollary in this paper has a companion machine-checked formal proof in the **Verum MSFS Corpus**, available at `https://github.com/gst-st/msfs`. The companion artefact ships its own README with the full paper-to-corpus map, the trusted-boundary report, the live audit surface (`verum audit —bundle` produces a one-command verdict), foreign-format exports (Coq, Lean 4, Agda, Dedukti, Metamath), and reproducibility instructions.

Trusted boundary. The MSFS verification is self-contained modulo:

- ZFC,

- two strongly inaccessible cardinals $\kappa_1 < \kappa_2$,

- the Verum trusted kernel (`crates/verum_kernel/` of the Verum source tree).

Nothing else is admitted. The boundary is enforced by the kernel-side invariant `verum_kernel::mechanisation_roadmap::msfs_self_contained()` and by the corpus-side Verum-language theorem `msfs_self_containment_theorem`; both are re-checked on every build and refuse any commit that introduces an external dependency within MSFS scope.

Citations to external authorities (Lurie HTT 2009, Adámek–Rosický 1994, Bergner–Lurie 2021, Riehl–Verity 2022, Whitehead, Pronk's bicategory-of-fractions, Lawvere's fixed-point theorem) appear in the corpus as `@framework(name, "citation")` markers and are enumerated by `verum audit —framework-axioms`.

Acknowledgments and disclosures

The author thanks the mathematical community for the foundational results of Cantor, Russell, Gödel, Tarski, Lawvere, Lambek–Scott, Adámek–Rosický, Makkai–Paré, Jacobs, Pronk, Lurie, Riehl–Verity, Voevodsky, Schreiber, and Clausen–Scholze, on which this synthesis rests. Max Sereda is the sole author; conceptualisation, formal analysis, proofs, and writing are entirely the author's responsibility. The work was conducted independently; the author declares no competing interests and no external funding. No data were generated or analysed; the full LaTeX source is archived with the preprint deposit.

References

[1] P. Aczel, *Non-Well-Founded Sets*, CSLI Lecture Notes 14, Stanford, 1988.

[2] J. Adámek and J. Rosický, *Locally Presentable and Accessible Categories*, London Mathematical Society Lecture Notes 189, Cambridge University Press, 1994.

[3] D. Ara and G. Maltsiniotis, *Homotopy comparison of cubical and globular models of weak ω-categories*, Preprint, 2017.

[4] S. Awodey and M. Warren, Homotopy theoretic models of identity types, *Mathematical Proceedings of the Cambridge Philosophical Society* 146 (2009), 45–55.

[5] I. Baković, *Fibrations of bicategories*, Preprint, 2010.

[6] C. Rezk, A Cartesian presentation of weak n-categories, *Geometry & Topology* 14 (2010), no. 1, 521–571; arXiv:0901.3602.

[7] J. Bénabou, Introduction to bicategories, in *Reports of the Midwest Category Seminar*, Lecture Notes in Mathematics 47, Springer, 1967, pp. 1–77.

[8] J.E. Bergner and C. Rezk, Comparison of models for (∞, n)-categories, I, *Geometry & Topology* 17 (2013), no. 4, 2163–2202.

[9] C. Barwick and C. Schommer-Pries, On the unicity of the theory of higher categories, *Journal of the American Mathematical Society* 34 (2021), 1011–1058; arXiv:1112.0040.

[10] G. Cantor, Über eine elementare Frage der Mannigfaltigkeitslehre, *Jahresbericht der Deutschen Mathematiker-Vereinigung* 1 (1891), 75–78.

[11] D. Clausen and P. Scholze, *Lectures on Condensed Mathematics*, Lecture notes, University of Bonn, 2019, `https://www.math.uni-bonn.de/people/scholze/Condensed.pdf`.

[12] M. Ernst, The prospects of unlimited category theory: doing what remains to be done, *Review of Symbolic Logic* 8 (2015), no. 2, 306–327.

[13] S. Feferman, Systems of predicative analysis, *Journal of Symbolic Logic* 29 (1964), no. 1, 1–30.

[14] M.A. Shulman, All $(\infty, 1)$-toposes have strict univalent universes, preprint, 2019; arXiv:1904.07004.

[15] A. Gagna, Y. Harpaz, and E. Lanari, Gray tensor products and lax functors of $(\infty, 2)$-categories, *Advances in Mathematics* 391 (2021), 107986; arXiv:2002.06055.

[16] N. Gambino and R. Garner, The identity type weak factorisation system, *Theoretical Computer Science* 409 (2008), 94–109.

[17] K. Gödel, Über formal unentscheidbare Sätze der Principia Mathematica und verwandter Systeme I, *Monatshefte für Mathematik und Physik* 38 (1931), 173–198.

[18] C. Hermida, Some properties of Fib as a fibred 2-category, *Journal of Pure and Applied Algebra* 134 (1999), 83–109.

[19] M. Hofmann, *Extensional concepts in intensional type theory*, PhD thesis, University of Edinburgh, 1995.

[20] J.M.E. Hyland, The effective topos, in *The L.E.J. Brouwer Centenary Symposium*, Studies in Logic and the Foundations of Mathematics 110, North-Holland, 1982, pp. 165–216.

[21] B. Jacobs, *Categorical Logic and Type Theory*, Studies in Logic 141, Elsevier, 1999.

[22] E. Bishop, *Foundations of Constructive Analysis*, McGraw-Hill, 1967.

[23] D.S. Bridges and L.S. Vita, *Techniques of Constructive Analysis*, Universitext, Springer, 2006.

[24] S.R. Buss, *Bounded Arithmetic*, Bibliopolis, Naples, 1986.

[25] A.S. Esenin-Volpin, Le programme ultra-intuitionniste des fondements des mathématiques, in: *Infinitistic Methods*, Pergamon Press / PWN, Warsaw, 1961, pp. 201–223.

[26] A.S. Esenin-Volpin, The ultra-intuitionistic criticism and the antitraditional program for foundations of mathematics, in: A. Kino, J. Myhill, R.E. Vesley (eds.), *Intuitionism and Proof Theory*, North-Holland, 1970, pp. 3–45.

[27] A.A. Markov and N.M. Nagornyj, *The Theory of Algorithms*, Mathematics and its Applications (Soviet Series), Kluwer, 1988 (English translation of Nauka, Moscow, 1984).

[28] S.G. Simpson, *Subsystems of Second Order Arithmetic*, 2nd ed., Cambridge University Press, 2009.

[29] J. Lambek and P.J. Scott, *Introduction to Higher-Order Categorical Logic*, Cambridge Studies in Advanced Mathematics 7, Cambridge University Press, 1986.

[30] K. Kapulkin and P.L. Lumsdaine, The simplicial model of univalent foundations (after Voevodsky), *Journal of the European Mathematical Society* 23 (2021), no. 6, 2071–2126; arXiv:1211.2851.

[31] C. Lair, Catégories modelables et catégories esquissables, *Diagrammes* 6 (1981), L1–L20.

[32] M. Makkai and R. Paré, *Accessible Categories: The Foundations of Categorical Model Theory*, Contemporary Mathematics 104, American Mathematical Society, 1989.

[33] G.M. Kelly, *Basic Concepts of Enriched Category Theory*, London Mathematical Society Lecture Notes 64, Cambridge University Press, 1982.

[34] G.M. Kelly and R. Street, Review of the elements of 2-categories, Lecture Notes in Mathematics 420, Springer, 1974, pp. 75–103.

[35] F.W. Lawvere, Adjointness in foundations, *Dialectica* 23 (1969), 281–296.

[36] J. Lurie, *Higher Topos Theory*, Annals of Mathematics Studies 170, Princeton University Press, 2009.

[37] J. Lurie, *Higher Algebra*, Preprint, 2017.

[38] J.-Y. Girard, Locus solum: from the rules of logic to the logic of rules, *Mathematical Structures in Computer Science* 11 (2001), no. 3, 301–506.

[39] G. Priest, *In Contradiction: A Study of the Transconsistent*, 2nd edition, Oxford University Press, 2006.

[40] D. Pronk, Etendues and stacks as bicategories of fractions, *Compositio Mathematica* 102 (1996), no. 3, 243–303.

[41] M. Rathjen, The realm of ordinal analysis, in *Sets and Proofs* (Leeds 1997), London Mathematical Society Lecture Notes 258, Cambridge University Press, 1999, pp. 219–279.

[42] E. Riehl and D. Verity, *Elements of ∞-Category Theory*, Cambridge Studies in Advanced Mathematics 194, Cambridge University Press, 2022.

[43] B. Russell, *The Principles of Mathematics*, Cambridge University Press, 1903.

[44] U. Schreiber, *Differential cohomology in a cohesive ∞-topos*, arXiv:1310.7930, 2013.

[45] M.A. Shulman, 2-toposes, *Journal of Pure and Applied Algebra* 210 (2008), 273–287.

[46] T. Streicher, *Semantics of type theory*, Birkhäuser Progress in Theoretical Computer Science, 1991.

[47] A. Tarski, Der Wahrheitsbegriff in den formalisierten Sprachen, *Studia Philosophica* 1 (1936), 261–405.

[48] The Univalent Foundations Program, *Homotopy Type Theory: Univalent Foundations of Mathematics*, Institute for Advanced Study, Princeton, 2013, `https://homotopytypetheory.org/book/`.

[49] V. Voevodsky, *Foundations* Coq library, software artefact, 2010–2015, `https://github.com/vladimirias/Foundations`.

[50] B. Werner, *Une Théorie des Constructions Inductives*, PhD thesis, Université Paris 7, 1994.

[51] N.S. Yanofsky, A universal approach to self-referential paradoxes, incompleteness and fixed points, *Bulletin of Symbolic Logic* 9 (2003), 362–386.